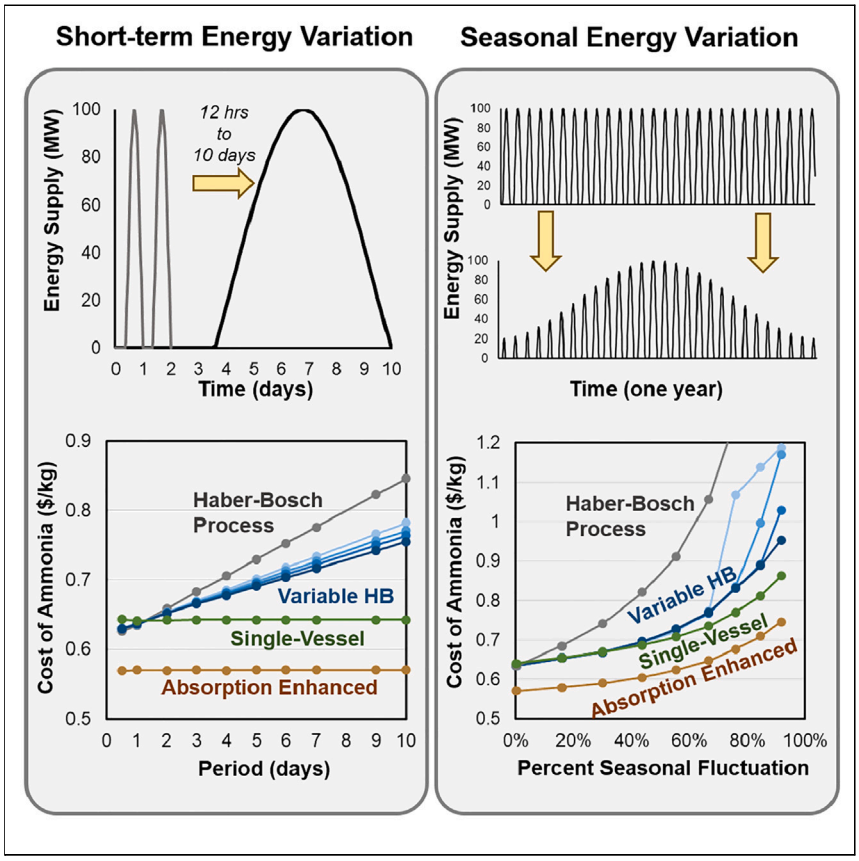
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The importance of dynamic operation and renewable energy source on the economic feasibility of green ammonia



Green ammonia produced from renewable electricity will enable net-zero by enabling sustainable fertilizer production and long-term energy storage. This work analyzes the effect of energy supply intermittency and scale on the economic feasibility of green ammonia production, highlighting the need for novel green ammonia production processes designed for low-pressure operation, agility, and reduced capital costs.

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Highlights

Assessing of daily and seasonal energy dynamics is essential for feasible economics

Conventional Haber-Bosch is hampered by long-term energy fluctuations

Novel fully flexible processes can adapt to intermittent energy supplies

At small scales, low-capital processes outcompete conventional Haber-Bosch

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The importance of dynamic operation and renewable energy source on the economic feasibility of green ammonia

Collin Smith[1] and Laura Torrente-Murciano[1,2,*]

SUMMARY

Long-term chemical storage of renewable energy in green ammonia has the potential to create a carbon-free energy market and replace fossil fuels. Cost estimations of green ammonia production are normally based on large-scale renewable energy installments with plentiful and consistent energy, which are well suited for the capital-intensive and flexibility-constrained Haber-Bosh process. This study analyzes green ammonia production under a wide range of scenarios by balancing the two competing economic drivers: the maximization of overall ammonia production and the minimization of feed buffer storage. We demonstrate that the optimized cost of green ammonia production (i.e., optimum scheduling, energy curtailment, and design capacity) is strongly dependent not only on the renewable energy capacity factor but also, more importantly, on the seasonal variations, production scale, and ramping capabilities of the process (i.e., dynamic operation). These results demonstrate the shift in process optimization target from maximizing energy efficiency to decreasing costs, which will be prevalent when electrifying chemical processes.

INTRODUCTION

The production of ammonia through the Haber-Bosch (HB) process for its use as fertilizer has been at the foundation of the fossil-fuel-dependent chemical industry for over a century. However, recent momentum has shifted toward green ammonia synthesized with renewable electricity, enabling both sustainable fertilizer production and long-term storage of renewable energy in its chemical bonds. In the latter case, ammonia is widely advantageous because it is a dense form of energy storage that is already stored cheaply and transported worldwide as a fertilizer.[1–3] Due to the challenges in operating modern energy systems with a high fraction of intermittent renewables,[4,5] ammonia storage is being explored to align production and demand as a replacement to conventional tactics for grid reliability such as long-range transmission[6] and as an alternative to other storage approaches such as compressed air,[7] batteries,[8,9] or hydrogen.[10–13] Accordingly, extraction of energy from ammonia through decomposing back to hydrogen as feed for proton exchange membrane (PEM) fuel cells,[14,15] direct ammonia fuel cells,[16] or combustion of ammonia in a turbine/engine[17] has become a flourishing topic of research.[18] Indeed, due to its prevalence as fertilizer in shipping markets, ammonia-powered engines for maritime transport are widely acknowledged to be achievable in the near future.[19]

The conventional fossil-fuel-powered HB process for ammonia synthesis has been highly optimized over the past century, culminating in a process that is capital-intensive,

CONTEXT & SCALE

Since the early 20th century, ammonia production has been linked to population growth through the use of synthetic fertilizer, but it contributes to ~2% of global carbon dioxide emissions. As we move toward net-zero, "green ammonia" produced using renewable electricity will be a pillar of a decarbonized chemical industry. At the same time, renewable electricity stored in the chemical bonds of ammonia will enable long-term storage and global transport of renewable energy using existing ammonia-handling infrastructure to replace global fossil-fuel markets.

To achieve this transition, the traditional paradigms of process design in the chemical industry need to be re-optimized for the novel constraints of renewable energy. This work highlights the effect of renewable energy intermittency and scale on ammonia production in terms of consequent economic costs, thereby demonstrating the perspectives required for green ammonia and further electrification of the chemical industry in the future.

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benefiting from the economy of scale, continuously operated, and heat integrated between the steps for hydrogen production and ammonia synthesis. However, when switching to renewable electricity to power the synthesis of green ammonia, hydrogen production takes place through electrolysis rather than fossil-fuel reforming, breaking its integration with the ammonia production step and inducing the need for a re-optimization of the process.[20] With the new framework of distributed and intermittent renewable energy, the optimization constraints shift to flexible and low-capital production processes. However, the HB process has limited operational flexibility to dynamically adjust production level and is capital-intensive as a result of high operating pressures (>100 bar). An alternative approach is to improve ammonia separation in the loop by using absorption in metal halides, which removes ammonia to lower concentrations than condensation, thereby facilitating a lower-pressure process (~20 bars) with increased flexibility for dynamic ramping of production level.[21,22] Furthermore, the ability of some absorbents to uptake ammonia at high temperatures has led to the development of integrated catalysis and separation of ammonia in a single vessel (S-V).[23] By removing the need for recycle and subsequent heat exchange, the S-V process notably reduces complexity, thereby easing capital requirements and ramping constraints. Even as the economic comparisons between the HB and absorption processes have emerged in recent years,[24,25] a fuller comparison is needed with respect to the effect of production scale and intermittency on cost while also including new technological innovations such as the novel S-V process.

While estimates in literature for the cost of green ammonia production span a wide range of values due to the uncertainty in the dominant process costs (i.e., renewable electricity, electrolyzer, and buffer storage), there is a general outlook that the production cost of green ammonia will approach that of brown (i.e., fossil-fuel-derived) ammonia in the near future.[26–28] However, current cost assessments have often used data from large-scale renewable energy installments with plentiful and consistent energy, well suited for the capital-intensive and flexibility-constrained HB process.[3,26] While it has also been indicated that the effect on cost of the overall capacity factor of the renewable energy and the ramping capabilities of the HB process can be significant, an investigation of the central relationship between the characteristic dynamics of renewable energy and the flexibility constraints of ammonia production has been lacking. Furthermore, the alignment of intermittent (renewable energy supply) and constrained (ammonia synthesis) processes with time-variable ammonia production and the consequent capital utilization has not been adequately studied.

Due to the variability of electricity supply from solar or wind energy, the optimal design of green ammonia production needs to consider both the maximum production rate (i.e., capacity) and the actual production at any given time (i.e., scheduling) for both hydrogen and ammonia production (HB). Such a complex system is typically solved using mixed integer linear or nonlinear equations with potential computational simplifications including compression of annual data into representative periods, linearization of discrete variables, and decoupling unit sizing with its subsequent scheduling.[1,28–31] Alternatively, several mathematical methods have been used for the determination of scheduling, with various ways of simplifying the time-variable ammonia production.[3,27,32] In this context, simplified methods are reasonable due to the uncertainty associated with the unit cost estimates. However, investigating the relationship between energy supply dynamics and production flexibility requires a method that directly balances the two competing economic drivers: the maximization of the overall ammonia production and the minimization of feed buffer storage.

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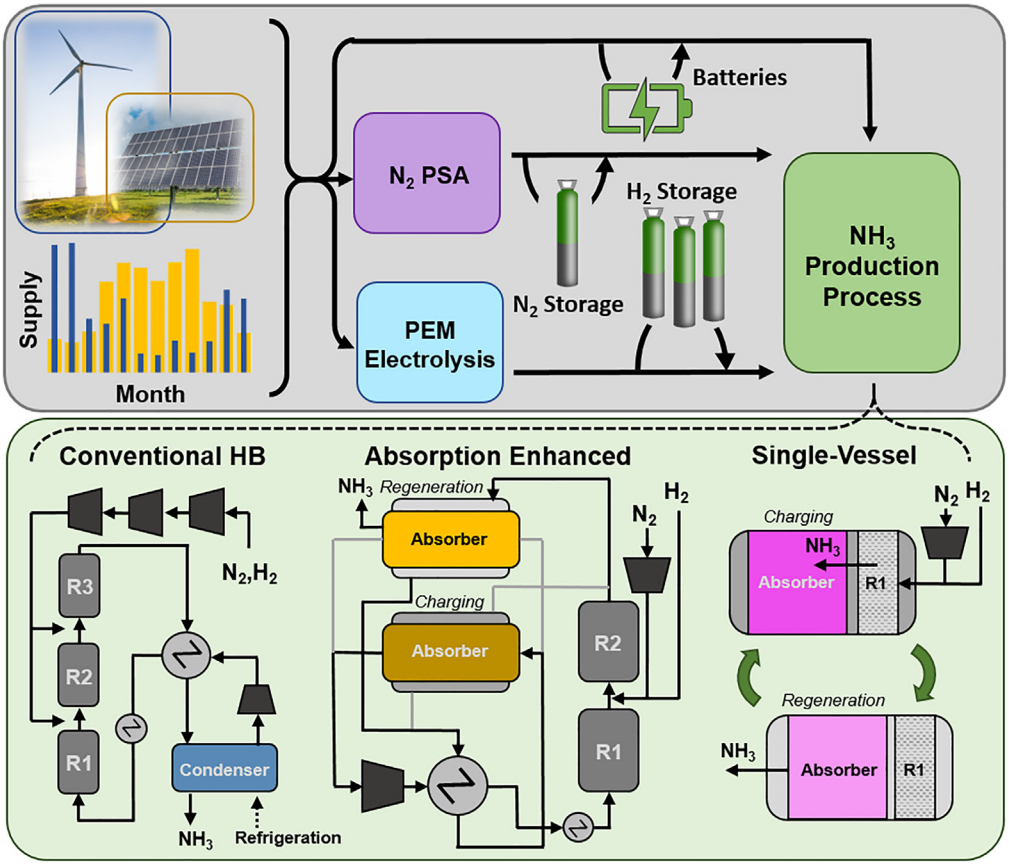


Figure 1. Schematic of green ammonia production from solar and wind energy

Ammonia is produced through the conventional HB process (with or without ramping capabilities), the absorption-enhanced process, or the single-vessel integrated process. When using the conventional HB process, batteries and gas storage are implemented to buffer fluctuations in renewable energy supply due to constrained ramping capabilities.

In this study, we present an economic assessment of the interrelationship between the dynamics of energy supply and the flexibility of ammonia production, taking into consideration the scale-cost dependency. This is accomplished using the system shown in Figure 1, where various energy sources are paired with available ammonia production technologies with standard hydrogen production, nitrogen production, and buffer storage processes in-between. The profiles of energy supply are first analyzed through a generalized model of two super-imposed sine waves to simulate short- and long-term variations and are further investigated in detail through case studies of solar and wind energy with a range of capacity factors and energy supply variability. Green ammonia production is modeled using a constant production HB process, a variable production HB process (i.e., with ramping capabilities), an absorption-enhanced (Abs.Enh) process (where condensation is replaced by absorption) or S-V integrated synthesis and absorption. The distinct abilities of these processes to dynamically adjust production level to follow intermittent renewable energy supply as well as scale-cost dependencies are highlighted in this analysis. It is demonstrated that the optimized cost of green ammonia production (i.e., optimize scheduling, energy curtailment, and design capacity) as a function of process flexibility (i.e., dynamic operation) is only moderately dependent on short-term energy variations but exponentially dependent on seasonal variations, where a fully flexible process becomes advantageous. These results provide useful

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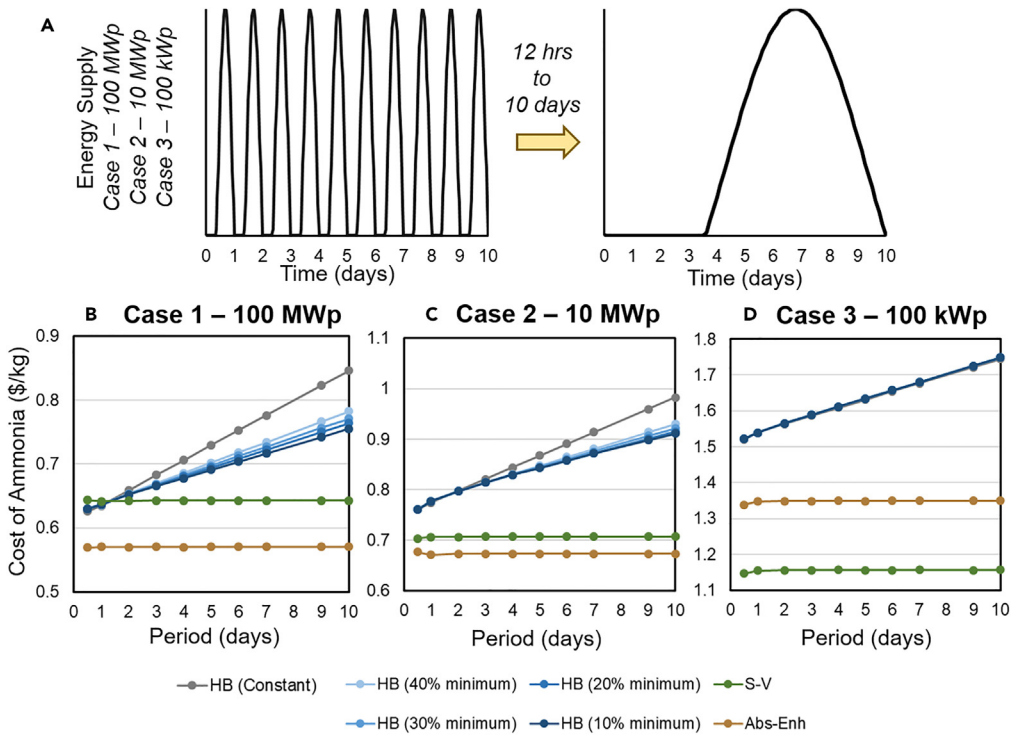


Figure 2. Isolated effects of short-term variations on the production cost of green ammonia using a 100 MW, 10 MW, and 100 kW peak renewable energy installations for cases 1, 2, and 3, respectively

(A) Schematic of model of the short-term variations as a cropped sine wave with variable period between 12 h and 10 days for the different peak capacities.

(B–D) Optimized production cost of green ammonia as a function of short-term variation period. A constant energy cost of 2.5 ¢/kWh and lineal scaling factors for cost estimates are assumed.

guidelines for the optimization of future green ammonia facilities, considering scale, dynamics of energy supply, and implementation of emerging synthetic technologies.

RESULTS AND DISCUSSION

Effect of short- and long-term fluctuations on cost of green ammonia

To evaluate the isolated effect of short-term and long-term energy supply variations on the cost of green ammonia production, the renewable energy supply was modeled as a cropped sine wave (see Figure 2). Peak energy supply was modeled in three cases as 100 MW, 10 MW, and 100 kW. To ensure periods of zero energy supply in the model (i.e., to mimic solar energy at night), the wave was shifted such that the bottom 30% of the energy supply oscillation was below zero, which was then set to zero for the purpose of analysis. To isolate the effect of energy intermittency from scale in this analysis, the cost of energy was held constant at 2.5 ¢/kWh and nonlinear capital cost relations were linearized at the average production capacity. Four different ammonia synthesis technologies were evaluated: (1) constant production HB process, (2) variable production HB process, (3) Abs.Enh loop process, and (4) S-V integrated process. For simplicity, it is assumed that the oscillations of the renewable energy input are constant (i.e., constant period) throughout the year, ranging from 12 h to 10 days, as depicted in Figure 2A. Fully flexible technologies that are able to quickly ramp up and down following these oscillations (such as the Abs.Enh and the S-V processes) are cost-insensitive to the variation of such short-term oscillations. However, the scale of the renewable energy installation has a

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strong effect on the cost as discussed in detail later. It is important to highlight that this is the optimized (i.e., minimum) cost of green ammonia production by optimizing scheduling and energy utilization. On the other hand, the cost of green ammonia produced using the HB process increases linearly as the oscillation period increases due to the need for buffers to cope with such short variations, and the magnitude of this increase is dependent on whether the process operates constantly or can ramp down to 10%, 20%, 30%, or 40% of its design capacity. The higher the flexibility range of the synthesis technology, the smaller the cost dependency on short-term renewable energy variation.

When the period of supply fluctuations is short (<2 days), the ramping ability of the HB process is insignificant, as shown in Figures 2B and 2C. As the duration of operating with an energy deficit is short, the hydrogen storage contributes to <2% of the overall ammonia production cost; therefore, it is optimal to keep the HB process at constant production such that its capacity factor is high. As the period of renewable energy supply increases >2 days, the hydrogen storage cost becomes dominant, with the process benefiting from a dynamic operation. Increasing the flexible capacity range from 40%–100% to 10%–100% minimizes the cost for hydrogen buffer, with a greater effect on the green ammonia cost as the period of renewable energy supply increases. Indeed, with a period of 10 days, the cost for hydrogen buffer increases to 14% and 19% of the overall cost in the cases of 10% and 40% minimum production, respectively (not shown). As expected from this analysis, the cost of green ammonia production via the constant production HB process is penalized by the cost associated with hydrogen storage, with the greatest linear increase with respect to the short-term period of all the studied technologies. Interestingly, at the small scale of 100 kW (Figure 2D), the increased cost of the HB process enables constant optimal production in all cases to maximize capital utilization.

Similarly, the isolated effect of long-term variations (i.e., seasonal variations) on the cost of green ammonia was evaluated by fixing the short-term oscillation period at 1 day and adding a second wave function (period of 1 year) subtracted from the first, whereby the effect of seasonal variations could be assessed by adjusting the amplitude of the second wave function, as shown schematically in Figure 3A. All the green ammonia production technologies, independent of their ramping capabilities, are detrimentally affected as the percentage of energy reduction in the lean months increases.

As shown in Figures 3B–3D, while there is minimal increase in cost with <20% energy reduction in lean months relative to peak months, the cost increases rapidly in an exponential manner with greater levels of long-term energy supply variations. The increase in green ammonia production cost is in part due to an overall lower capital utilization associated with the lower overall energy supply. For the flexible production processes (S-V and absorption-enhanced processes), the increase in cost is considerably less sharp than the HB processes because they benefit from low capital, particularly at small scales. At high seasonal variations (>70%), the benefit of fully flexible ramping processes becomes even more pronounced because the processes can turn on/off in low-energy months to utilize that energy supply without the need for storage. As a result, as the energy supply becomes non-ideal (greater seasonal variation) novel, agile, and flexible ammonia production processes become critical.

The constant production HB process is unable to cope with such seasonal variations in an economically feasible manner, being greatly penalized by the need for large-scale seasonal hydrogen storage. With ramping capabilities implemented in the

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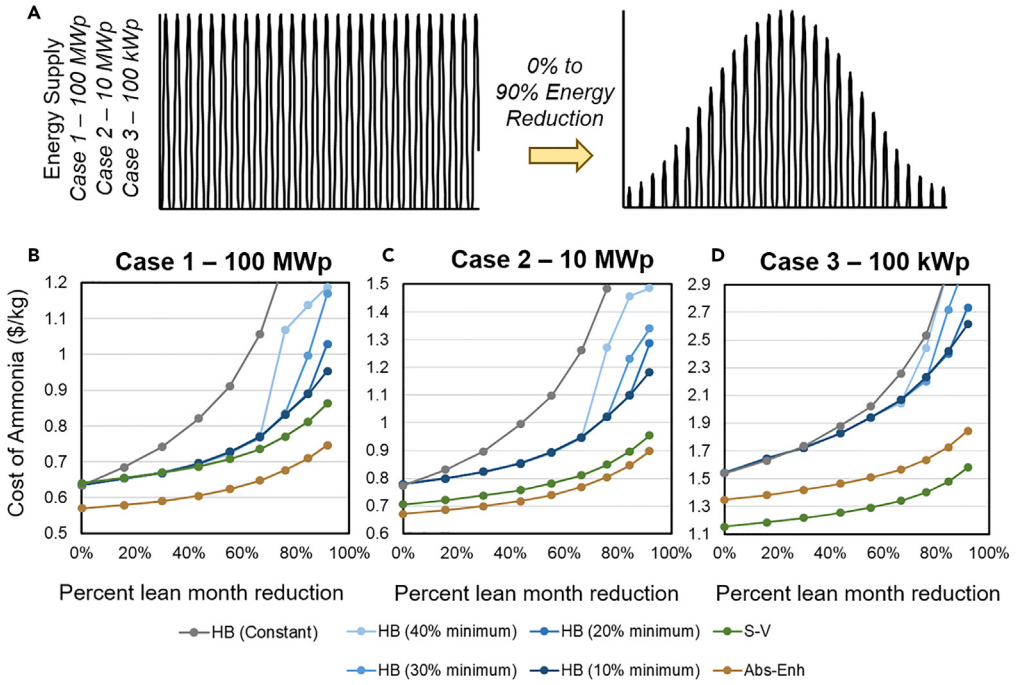


Figure 3. Isolated effects of long-term variations on the production cost of green ammonia using a 100 MW, 10 MW, and 100 kW peak renewable energy installations for cases 1, 2, and 3, respectively

(A) Schematic of model of the long-term variations representing seasonal variations as a sine wave function with variable period subtracted from a cropped initial sine wave with constant 1-day period (note: short-term fluctuations not to scale).

(B–D) Optimized production cost of green ammonia as a function of fraction reduction in energy during lean months (relative to peak months). In simulations of with the ramping HB process, the scheduling option for 7 days shutdowns was provided. A constant energy cost of 2.5 ¢/kWh and lineal scaling factors for cost estimates are assumed.

HB process, production can be reduced in lean months to mitigate hydrogen storage, resulting in a less sharp increase in ammonia production cost. When the percentage in energy reduction in lean months is <70%, there is no benefit in increasing the range of production by altering the lower limit from 40% to 10% because the process is not required to ramp down to the lower limits to cope with such seasonal variations. However, when the seasonal energy supply reduction is >70%, the minimum production level becomes significant because it triggers the trade-off between increased hydrogen storage to keep the process operational and decreased energy utilization if the process shuts down. As such, a HB process with 40%–100% capacity range will be hampered by this trade-off at seasonal energy reductions of >70%, while a capacity range of 10%–100% will only be hampered at seasonal energy reductions of >90%.

To delve deeper into the details of the economic trends using this renewable energy supply profile with a sine wave model, three case studies of solar and wind energy are selected with a range of characteristic dynamics: solar and wind energy in Cambridge, UK, and solar energy in Madrid, Spain. Solar energy in Madrid is characterized by being plentiful (i.e., high-capacity factor of 0.21) with small seasonal variations (~40% energy reduction in winter) and well-defined daily periodic fluctuations. Solar energy in Cambridge has not only a smaller capacity factor (0.1) but also increased seasonal energy reduction in the winter months (~75%). Both of them have well-defined short-term variation periods of 24 h. Wind energy in Cambridge has a high-capacity factor (0.26) with a ~3-day period and a 90% reduction in summer months. The detailed analysis of these case studies further supports the

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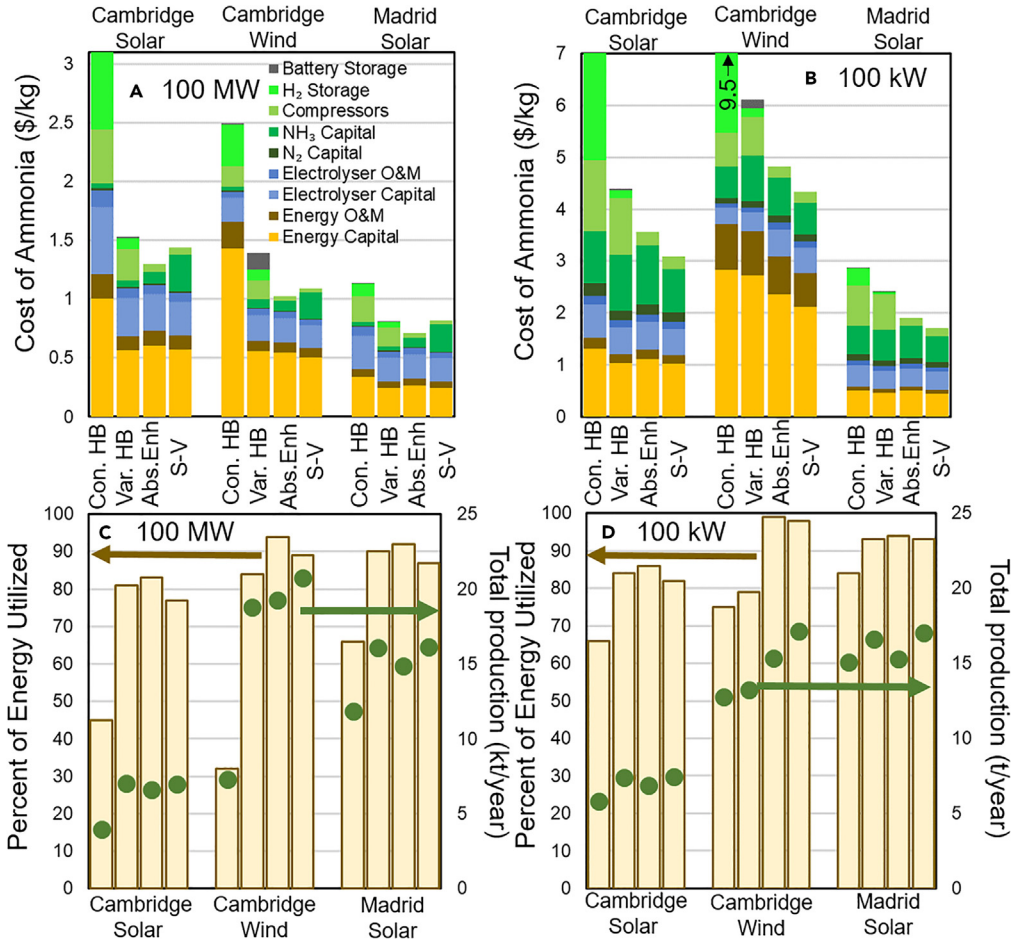


Figure 4. Economics of ammonia production as a function of ammonia production process and energy source

Ammonia is produced using the constant Haber-Bosch (con. HB), variable Haber-Bosch (var. HB), absorption-enhanced (Abs.Enh), and single-vessel (S-V) processes with solar and wind energy in Cambridge and solar energy in Madrid.

(A and B) Optimized cost breakdown for (A) 100 MW and (B) 100 kW renewable energy installation.

(C and D) Energy utilization and total production for (C) 100 MW and (D) 100 kW renewable energy installation. Operation and maintenance (O&M) costs are included for the electrolyzers and renewable energy installment.

generalized analysis of ammonia production processes paired with short-term and seasonal energy variations.

Cost breakdown and energy utilization

The critical features in the cost of green ammonia production using exclusively renewable energy (in a truly off-grid and off-pipe situation) are illustrated in Figure 4 for three case studies of renewable energy supplies (wind and solar energies in Cambridge and solar energy in Madrid) at two extreme scales (100 MW and 100 kW peak capacity installations). It should be noted that the production scale at 100 MW is an order of magnitude less than the current brown ammonia production, but it is anticipated that future green ammonia production will occur at a smaller scale. Across all the case studies, the capital costs of renewable electricity and electrolyzers are the dominant process costs (~70%–80% of cost at 100 MW), with significant buffer storage when ammonia is produced through the constant production HB process, in agreement with the pre-existing consensus. When using a large-scale energy supply with a high-capacity factor and minimal seasonal variation—represented by solar energy in Madrid—the cost of green ammonia production is on par with existing

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knowledge. With a small-scale energy supply (100 kW) and non-ideal characteristics (i.e., solar and wind energy in Cambridge), the cost breakdown highlights the benefit of flexible and low-capital ammonia production, as discussed below.

The use of the constant production HB process consistently results in higher green ammonia production costs (Figures 4A and 4B) due to its strong dependence on hydrogen storage to level seasonal fluctuations. At large scales, the use of flexible processes (variable HB [Var. HB] with 10% minimum production, Abs.Enh, and S-V) to accommodate intermittencies in renewable energy results in small differences in overall cost. The ability to ramp production reduces or eliminates the need for buffer storage, and the cost for compressors in the Var. HB process (~15% of total cost) is balanced by the high cost of repeat ammonia production modules in the S-V process (~20% of total cost) without economies of scale. The compressor and ammonia loop costs are the lowest for the Abs.Enh process because it retains economies of scale while also operating at a lower pressure. However, this is partially balanced by an increased energy cost due to inefficiencies in absorber pressure drop and regeneration that results in less ammonia production even while utilizing more energy (Figure 4C).

At the limit of small scales (100 kW), although the cost universally increases due to the higher cost of solar panels, wind turbines, and electrolyzers, among the production processes, it is clear that the S-V process is superior due to its simplicity and modular nature. When using HB-based ammonia production technologies, including the Abs.Enh process, there is a substantial higher cost of process equipment (NH_3 capital and compressors), due to the lack of well-known economies of scale for this process, thereby highlighting the challenge of distributed manufacturing with these processes. Indeed, it is likely that the current cost estimates underestimate the cost of implementing the HB process at small scales (e.g., compressors and heat exchangers), while the design of the S-V process is suited for reliable estimates at a small scale. Considering these estimation limitations, the cost of the ammonia synthesis at small scales (green components in Figure 4B, including H_2 storage) accounts for ~54%–60% of the production cost when using a constant production HB process and 20%–44% when using the fully flexible processes. The range of the values depends on the source of energy used.

Energy utilization, defined as the percent of supplied energy used in ammonia production and shown in Figures 4C and 4D, is a crucial process parameter when comparing constant and variable production. All available energy is assumed to be purchased (through capital and operating cost of panels/turbines); therefore, the energy not utilized is curtailed. When ammonia is produced through constant capacity HB at 100 MW, energy utilization is low because it is more cost effective to curtail the energy rather than store over the seasons as the magnitude of corresponding storage requirement would be high and thus expensive. However, at a small scale (100 kW), the increased cost of producing energy at this stage shifts the optimization toward including the necessary storage and decreasing curtailment. At both scales, variable ammonia production results in an increased energy utilization (>80%). Curtailing low percentages of energy is a primary factor in decreasing the cost of ammonia with these technologies, and it is expected that combining solar and wind energy will further reduce curtailment by leveling seasonal variation and thereby decrease cost. Nevertheless, the balance between energy utilization and economic optimization illustrated here is contrary to the traditional process optimization and is expected to become a feature in the electrification of the chemical industry.

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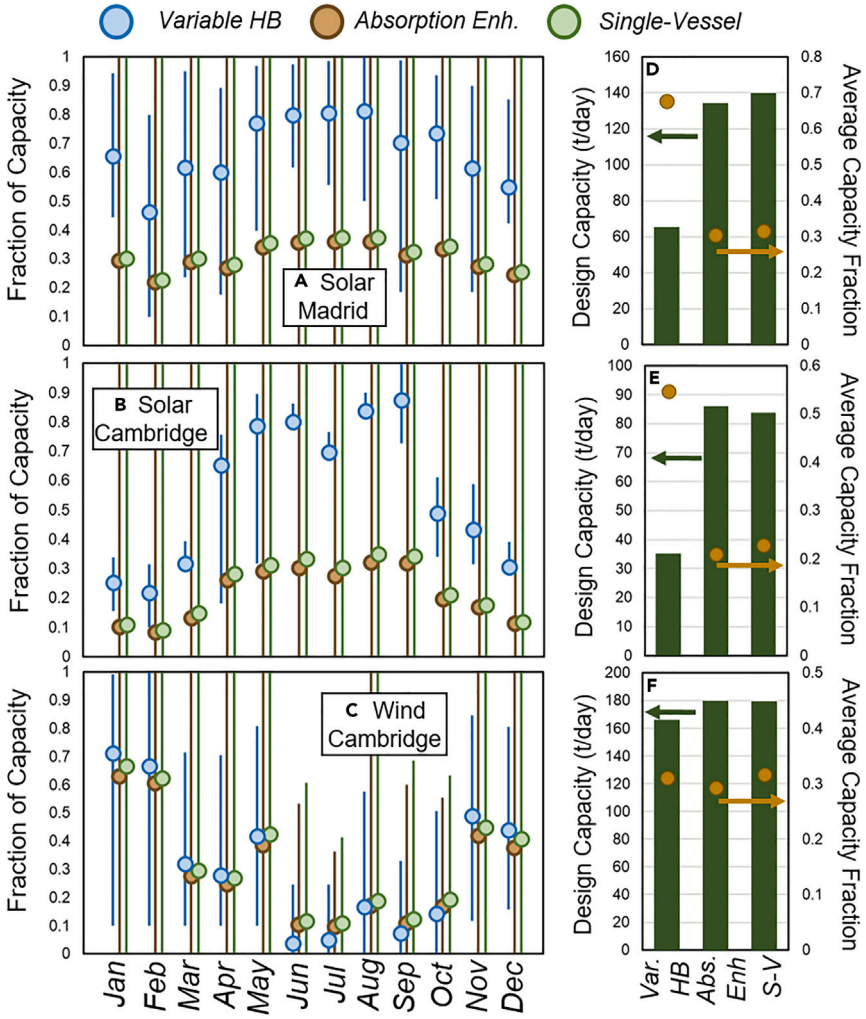


Figure 5. Dynamics of ammonia production over a year using 100 MW renewable energy installations

(A–C) The average (circle) and range (line) of fractional capacity in each month with the variable HB, absorption-enhanced, and single-vessel processes for solar energy in Madrid (A), solar energy in Cambridge (B), and wind energy in Cambridge (C).

(D–F) Design capacity and average yearly capacity fraction as a function of ammonia production process.

Evaluation of the dynamic ammonia production

The response of the different ammonia production technologies with ramping capabilities to the transient energy supply and its effect on the economics of ammonia production are evaluated as a function of energy source in the case studies. Figure 5 presents the optimized dynamic operation of the different ammonia production technologies over a representative year using 100 MW renewable energy installations. The three technologies accommodate the intermittencies of renewable energy supply by varying their capacity fractions (represented by the lines), with the average capacities for each month represented by the circles (Figures 5A–5C).

Solar energy (Madrid)—Short-period fluctuations, small seasonal variations

Solar energy in Madrid is characterized by being plentiful (i.e., high-capacity factors) with small seasonal variations and well-defined daily periodic fluctuations. In this case, there is little differentiation in the resulting optimized cost of green ammonia production among the three ammonia production processes with dynamic

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operation (Figure 4A) while leading to similar annual productions (Figure 4C). Due to the small seasonal variations, the period of supply oscillations is shorter than that critical to the variable production HB process, thereby the advantages of the higher and faster ramping capabilities of the Abs.Enh and S-V integrated processes are not fully exploited. As indicated by the range of capacity for each month of the year (Figure 5A), the ammonia production in the fully flexible processes (Abs.Enh and S-V) follows the energy supply availability, stopping production when renewable energy is not available (i.e., during night) and consequently presenting a low rate of capital utilization (~0.3 average capacity fraction, Figure 5D). Conversely, the intrinsic need for buffer capacity of the variable production HB process (able to ramp down only to 10% capacity) requires a lower design capacity of the ammonia production loop (~65 t/day versus the ~130–140 t/day of the more flexible ammonia synthesis technologies). This is translated to a considerably higher ammonia process capital cost utilization (~0.68 average capacity fraction) as the production distributes along the day. These data demonstrate that the cost of compressors for high-pressure hydrogen storage to distribute production overnight in the Var. HB process offsets the reduction in costs for ammonia production due to high utilization, leading to similar cost of ammonia across all three processes (Figure 4A).

Solar energy (Cambridge)—Short-period fluctuations, large seasonal variations

Solar energy in Cambridge, despite being characterized by small capacity factors and seasonal energy reduction in the wind months, further highlights the effect of ammonia production technology on design capacity and the utilization of the ammonia loop capital indicated for solar energy in Madrid. An energy supply with non-ideal characteristics leads not only to a considerably lower annual production (~7 kt/year versus ~15 kt/year using solar energy in Madrid) but also a substantial increase in optimized green ammonia production cost (Figure 4A). The larger seasonal variations generate a strain on the variable production HB process, leading to an increased buffer hydrogen storage requirement, but the storage capacity that keeps the process operational during the winter months is used during the summer to maintain production overnight at a higher capacity factor than with solar energy in Madrid (Figure 5B). As a result, while seasonal fluctuations and lower capacity factor of the renewable energy sources decrease the overall capacity factor of the perfectly flexible ammonia production processes by 1/3 compared with solar energy in Madrid, the overall capacity factor of the Var. HB process only reduces by ~15% because buffer storage allows to keep fractional capacity high.

Wind energy (Cambridge)—Long-period fluctuations, large seasonal variation

The limitations of using an ammonia synthesis process with constrained ramping capabilities (i.e., variable production HB) are further exposed when using non-ideal renewable energy inputs with large short-term and seasonal variations such as wind energy in Cambridge. Indeed, in these cases, the full ramping flexibility of the Abs.Enh HB and the S-V integrated processes is fully exploited, resulting in not only lower optimized green ammonia production cost but also higher annual production and higher energy utilization (Figure 4). In addition to the ramping flexibility, the superior agility of the S-V integrated process to cope with the short-term variations and the greater energy efficiency of the process allows a higher annual production, following the trend of S-V > Abs.Enh > Var. HB. Interestingly, the three processes present a similar average capital utilization for the ammonia production process (Figure 5F) because the benefit of a buffer with the Var. HB process is limited due to extended periods without energy supply, particularly in the summer months.

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The limited ramping flexibility of the variable production HB process (i.e., able to ramp down to a minimum capacity of 10%) causes severe constraints (and high costs) because it must balance the hydrogen storage required to maintain minimum production in periods without energy against the increase of design capacity to utilize surplus energy. During winter months with plentiful energy supply, there are still extended periods with little energy produced (Figure S1), which induces a high design capacity to facilitate the ramp-up of the process during high peak energy supply (Figure 5C), resulting in an over-sized process. At the same time, with a 10% lower-capacity limit, the process struggles to accommodate low levels of energy supply in summer, needing to completely shut down the ammonia production over days and thus producing noticeably less ammonia than the fully flexible processes during 5 months a year (Figure 5C). In addition, it also requires excessive battery storage (Figure 4A) due to the inefficiency of the process at very-low-capacity levels. Therefore, in such a situation where the intransience of the conventional HB process is being strained by energy supply oscillations with a large period and/or high magnitude, as with solar and particularly wind energy in Cambridge, a perfectly flexible process finds greater benefit. While the Var. HB process is suitable for large-scale installations of consistent energy supply, novel processes with greater flexibility, such as the S-V integrated process, are needed to cope with non-ideal variations in renewable energy production even if they present high average capacities.

In addition, it should be noted that for all ammonia production processes, the fractional capacity of the hydrogen/nitrogen production is low (20%–30%) and shows high variability over the months as the gas production follows peaks in energy supply (Figure S8). Thus, although the variable production HB process has greater utilization of the capital associated with the ammonia synthesis loop than the other fully flexible technologies, the greatest capital costs (i.e., electrolyzers and energy supply) have a universally poor capital utilization of only 20%–30% for the three studied technologies, which is a direct consequence of needing to follow both daily and seasonal variations in energy supply.

Effect of ramping capability

In the previous case study analyses, we have considered a variable production HB process capable of ramping down to 10% of its design capacity in a relatively fast manner by varying the nitrogen/hydrogen ratio, except in the range of 10%–30% capacity when the ramping rate is limited by the rate at which reactants are fed into the system. However, these assumptions represent the best possible scenario, clearly differing to typical and likely more realistic scenarios in literature.[26,28] Indeed, it is well known in the industry that operating the HB process at low capacities could potentially lead to unstable and even unsafe operation. To evaluate the effects of more constrained operation of the HB process, Figure 6 compares the best case scenario (ramping capabilities between 10% and 100% capacity with minimum 3-day period for shutdown) with more realistic scenarios of 40% lower limit with 3-day shutdowns and 40% lower limit with 7-day shutdowns (i.e., a full month due to representative weeks). It is important to mention that, in this study, we are assuming a perfect forecasting of the renewable energy supply to schedule shutdown periods; however, difficulties predicting weather (especially relevant for wind energy resources) would likely make longer shutdown periods and higher minimum production more practical to act as a margin of error to avoid operating under a major energy deficit when forecasted energy supply is not generated.

When using plentiful and consistent renewable energy resources, such as solar energy in Madrid, with short-period oscillations, low seasonal variations, and high-capacity factors, the more flexible HB process (best case scenario) presents little advantages versus

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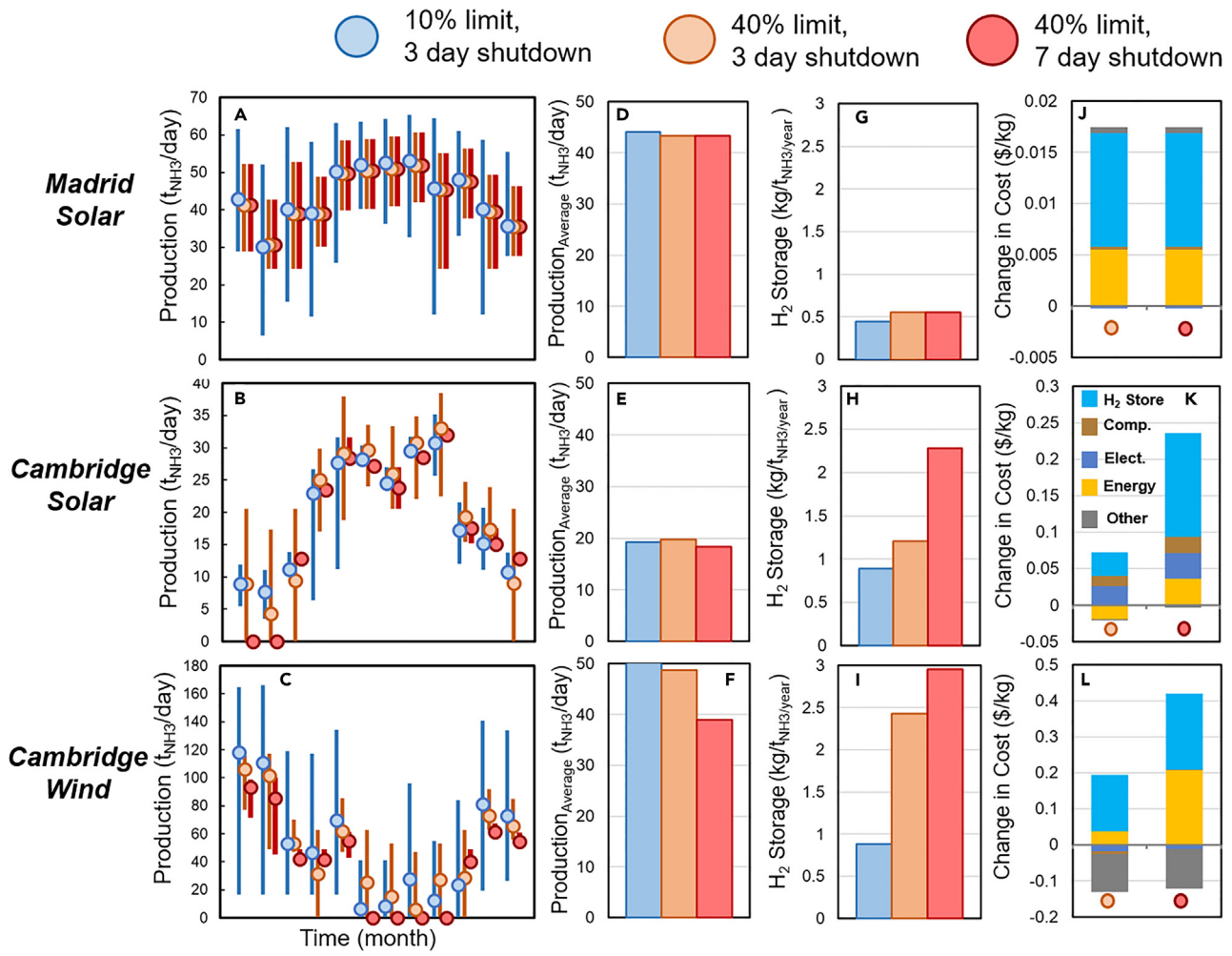


Figure 6. Effect of constraints on HB ramping at minimum capacity using a 100 MW renewable energy installation: comparison of 10% and 40% minimum capacity for 3 and 7 days minimum shutdown periods

(A–C) Average (circle) and range (line) of ammonia production for each month of the year for solar energy in Madrid (A), solar energy in Cambridge (B), and wind energy in Cambridge (C).

(D–F) Average rate of ammonia production over the course of a year.

(G–I) Hydrogen storage normalized by the total ammonia produced as a function of constraints on minimum capacity.

(J and K) Change in costs relative to the base case of 10% lower limit and 3 days shutdown period, sub-divided into the components of H_2 storage, compressors, electrolyzers, solar/wind, and other costs.

the more constrained HB processes. The three HB-based scenarios present similar average monthly production (Figure 6A), with slightly higher average capacity fraction for the less flexible process (0.7 rather than 0.67) due to less ramping oscillations. The increase in hydrogen storage requirement for the less flexible processes is small (Figure 6G) with its associated cost similarly minor (Figure 6J). As a result, the increase in the green ammonia cost is minimal (~0.017 $/kg), resulting from small increases in associated energy costs because less ammonia is made for the same solar energy capital investment. The overall annual production (Figure 6D) decreases slightly because the lower capacity limit places constraints on the maximum design capacity, which causes less overall energy utilization.

When applied to non-ideal energy sources (i.e., solar and wind energy in Cambridge with large seasonal variations), the economics are notably more sensitive to an

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increase in the lower limit of the HB process. Increasing the lower limit capacity from 10% to 40% (keeping constant the minimum shutdown period to 3 days) increases the green ammonia cost by 0.05 and 0.1 $/kg using solar and wind energy in Cambridge, respectively. The increment in cost is even more substantial if the shutdown period increases to a minimum of 7 days (at a lower limit capacity of 40%) up by 0.24 and 0.41 $/kg using solar and wind energy in Cambridge, respectively. It is important to note that this is above the baseline variable production HB process with 10% lower limit capacity, already more expensive than the fully flexible Abs.Enh HB and the S-V integrated processes in some cases.

When using low capacity and large seasonal variations renewable energy (i.e., solar energy in Cambridge), for a variable production HB process with a 40% lower limit capacity, it is optimal for the process to shut down for a period during the winter months when energy supply is low (Figure 6B). When the minimum shutdown period is 3 days, this is translated in shutdown periods over 4 different months. Extending the shutdown period causes no production in January and February, whereas maintaining minimum production in March and December to preserve overall energy utilization as much as possible. Although, the requirements for hydrogen storage increase considerably (Figure 6H), being more than double in the more restrictive case. As a result, production capacity consequently increases during more plentiful months to maximize energy utilization during the peak energy production season (thereby increasing electrolyzer and compressor cost).

Similarly, when a higher minimum capacity is imposed to the variable production HB processes driven by seasonally variable wind energy, the process is inevitably strained due to the aforementioned drastic fluctuation in energy supply between winter and summer. When the minimum shutdown period is 3 days, shutdown periods over 6 months are required to minimize the ammonia cost. Further increases to the minimum shutdown period to 7 days result in no production for 4 months per year because the average energy supply over the summer months is well below that required to maintain production at 40% of design capacity (Figure 6C). Hydrogen storage requirements increase considerably by at least ×3 in both cases (Figure 6I), and energy costs increase when the energy of summer months goes unutilized due to shutdown, translating into considerably higher green ammonia costs.

Thus, an analysis of Var. HB processes with greater constraints on ramping at minimum capacity further confirms the conclusion that alternative production processes with complete flexibility become advantageous with non-ideal energy sources that experience supply fluctuations of longer period and greater magnitude.

Conclusions

While the traditional HB process for the production of ammonia using fossil fuels has been designed and optimized to operate in steady state at full capacity to benefit from the economy of scale, such optimization fails when intermittent and variable renewable energy is used to produce green ammonia. Indeed, a careful assessment of the daily and seasonal variations of renewable energy supply as well as their capacity factor is key for the optimization of green ammonia processes, with each source requiring a tailored solution for process design and scheduling. The relationships presented here between the cost of ammonia with the short- and long-term variations provide guidelines for decision-making processes. Although capital cost for renewable energy and electrolyzers are still the dominant cost factor, small-scale installations and energy supplies with long-period and high-magnitude fluctuations

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require novel low-capital and flexible production processes, with ramping capabilities between 0% and 100% design capacity to warranty their economic feasibility.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Laura Torrente-Murciano (lt416@cam.ac.uk).

Materials availability

This study did not generate new unique materials.

Data and code availability

The datasets generated during this study are available at Apollo, the University of Cambridge data repository. DOI: https://doi.org/10.17863/CAM.104175.

Computational methodology

The economic calculations (capital, operational, and total ammonia costs) of ammonia production using a range of renewable energy sources were carried out by building a MATLAB algorithm. In all cases, both batteries and hydrogen/nitrogen storage (at 200 bar) were considered to buffer the intermittency of the renewable energy availability.

Renewable energy sources

To evaluate the effects of short-term and long-term variations in the renewable energy supply, a general model of the supply profiles was built by superimposing two sine waves, where the period of the primary sine wave and the amplitude of the secondary sine wave represent the short-term and seasonal energy fluctuations, respectively. To bolster the generalized model analysis, detailed case studies were conducted with three different sources of renewable energy to reflect varied costs, capacity factors, and short- and long-term intermittencies, as provided by the European Commission Photovoltaic Geographical Information System (PVGIS) with optimized fixed axes for solar panels[33] (illustrated in Figure S1). Solar energy in Cambridge has both a low overall capacity fraction (10%) and high variability between seasons, producing an average electricity cost of 5.3 ¢/kWh at 100 MW. Wind energy in Cambridge has a higher overall capacity factor (26%) with high installment cost, resulting in an electricity cost of 5.2 ¢/kWh at 100 MW, but produces dramatically more energy in winter than in summer. Solar energy in Madrid has both a high utilization rate (21%), with a cost of 2.7 ¢/kWh at 100 MW, and very little fluctuation between seasons. The energy costs were calculated from the total energy supply and the capital and operating costs estimates for solar and wind energy.[34] Each of the energy sources with an hourly resolution for 1 year was simplified by taking 1 random week for each month as representative and repeating them either four or five times depending on the month. While these case studies do not represent all possible solar/wind sources, they are instructive for highlighting the effect of supply intermittencies on the production cost of green ammonia as a function of production process dynamic operation.

Ammonia production processes

Four different processes were considered for the production of green ammonia: the conventional HB process with and without production ramping, the Abs.Enh process, and a S-V integrated processes, as shown in Figure 1. A foundational understanding of the difference between the ammonia production processes starts with

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modeling the processes and estimating the capital costs over a range of scales (as described in detail in the supplemental information). For that, the conventional HB loop process at 200 bar was construed as a three-stage adiabatic reactor using a conventional industrial iron catalyst (kinetic rate from Sehested et al.[35]), with minimum and maximum operational temperatures of 400°C and 500°C, respectively, and the inter-reactor quench cooling provided by fresh feed and ammonia separation through condensation at 10°C (Figure S2).

When the process is modeled at constant operation with production at 100% capacity, the reactors achieve the maximum temperature of 500°C at the outlet, with intercooling leading to the highest single-pass conversion through the reactor series. When the process is modeled for variable production, it has been simulated to be ramped up by adjusting the nitrogen/hydrogen ratio,[36] which is fundamentally similar to the method patented by Casale whereby an increased concentration of inerts decreases the production rate.[37] By using the nitrogen/hydrogen ratio to adjust the rate and equilibrium, it is simulated that the production rate can be rapidly ramped except at very low production capacities (10%–30%) where the rate of change is limited by the rate at which gas can be fed to the system (details in the supplemental information). In addition, while operating moderately below design capacity (40%–100%) only marginally increases the energy consumption associated with an increased recycle, at low production capacities (10%–30%), the energy consumption increases drastically due to failure of the heat integration between the inlet and outlet streams of the reactor (Figure S3). While 10% production capacity has been indicated to be achievable, it is also considered to be an optimistic target; therefore, lower production limits of 20%, 30%, and 40% are also being investigated.

The Abs.Enh process was modeled similar to the conventional HB process loop (same catalyst), with the condenser replaced by an absorbent (e.g., calcium chloride) that is able to remove approximately all ammonia produced by the reactor at 50°C. In this case, a lower pressure of 30 bar is used (the readily available pressure of electrolysis)[21,38] because absorption enables ammonia separation at considerably lower partial pressures.[22] A two-stage reactor with intercooling is used in this case because the equilibrium conversion is lower at 30 bar. As a result, this process can be safely ramped down by adjusting the recycle rate because the reactor reaches equilibrium before exceeding the maximum temperature of 500°C. While this adds practical complications involved with controlling inlet flow to match production and avoid overcooling the reactor feed, it allows the ability to ramp much faster than the conventional HB process and is assumed to be faster than the energy supply variations on the timescale of hours. The model considers two absorbers, one in operation, while the other one is regenerated at 300°C. Heat integration occurs through (1) the reactor effluent discharging its latent heat down to 300°C to the regenerating absorber, and (2) the remaining heat is transferred to the cool recycle. Compared with absorbers that lack heat integration, those with the heat integration will require practical considerations for controlling heat transfer in each step, but energy consumption is reduced by 10 kJ/mol_{NH3}. In addition, unlike in the conventional HB process, in the Abs.Enh process, the recycle compression energy constitutes a large component, especially at high production capacities (Figure S4) due to pressure drop in the absorber.

The S-V integrated process was modeled as two semi-batch reactors in parallel, containing a catalyst and an absorbent, one producing ammonia and the other one having the absorbent regenerated. During ammonia production, the reactor operates at ~330°C and 30 bar. The *in situ* separation of ammonia shifts the equilibrium, and the reactor is able to achieve close to 100% conversion without the need of recycle. The

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process was modeled using experimental kinetic data of a ruthenium/cesium/ceria catalyst and manganese chloride/silica gel absorbent because that is the available experimental data, with the assumption that the vessel is well mixed and negligible ammonia is decomposed during absorbent regeneration.[23] In contrast to the conventional HB process, this process can be ramped down completely (from 0% to 100% capacity) by simply decreasing the feed flowrate, allowing the pressure to drop until the ammonia pressure from reaction is no longer high enough to induce absorption. As a result of operating in a S-V, the heat of this process can be more easily integrated between the operating reactor and the regeneration step while negating the need of recycle compressors. As a result, the energy requirements for regeneration drop by 90% (increasing efficiency to ~81%), thereby putting the total energy requirement below that of both the conventional HB process (78% efficiency) and the Abs.Enh processes (67% efficiency, Figure S5).

Estimation of capital and operation costs

The total cost of green ammonia production was calculated using estimates for capital and operating costs of the major equipment in the system (listed in supplemental information), annualizing capital costs to a yearly basis using a 30-year lifetime and 7% discount rate and dividing by the total production. To isolate the effect of flexibility in ammonia production, synthesis gas production and storage units were approximated to have no flexibility constraints (i.e., able to ramp from zero to full capacity). Hydrogen and nitrogen production were modeled with PEM electrolyzers and pressure swing adsorption (PSA), respectively, with load-independent energy consumption. Buffer storage in lithium-ion batteries was considered to power the HB loop, and compressed hydrogen and nitrogen gases in steel cylinders at 300 bar provided feed gas, which cycle down to 200 bar for the HB process.[39] Crucially, the gas and battery buffer storage is not required for the Abs.Enh and S-V processes due to their fully flexible ramping capabilities.

Optimization of ammonia cost

The cost (in $/kg$_{NH3}$) for the production of ammonia with a series of specially formulated optimization algorithms in MATLAB. For the constant HB, Abs.Enh, and S-V ammonia production processes, optimization was conducted through adjusting the design capacity, with the scheduling following directly from the constraints of the process. In the case of the Abs.Enh and S-V processes with perfect ramping capabilities, the production of ammonia (and hydrogen) at every time point directly mirrors the energy supply up to the designed production capacity without the need for hydrogen/battery buffers. In the case of the constant production HB process, ammonia production was constant at the design capacity, with hydrogen (and nitrogen) production capacity optimized to balance decreased buffer storage and increased capital costs at higher hydrogen/nitrogen production capacity.

For the variable production HB process with ramping capabilities, the cost of ammonia was minimized using a "surplus-deficit" method, as shown schematically in Figure 7, to generate a suitable approximate solution for comparison purposes. The program initializes the scheduling and capacity of ammonia production at a constant value for each month corresponding to ~75% of the average energy supply for that month, adjusted over several optimization runs to find minimum cost of ammonia production (initialization has <5% effect on final cost). After initialization, the method adjusts ammonia production by increasing it during periods of surplus energy and decreasing it during periods of energy deficit, thereby simultaneously tuning production capacity and scheduling parameters. At the same time, within each iteration of tuning ammonia production, the capacity of hydrogen (and

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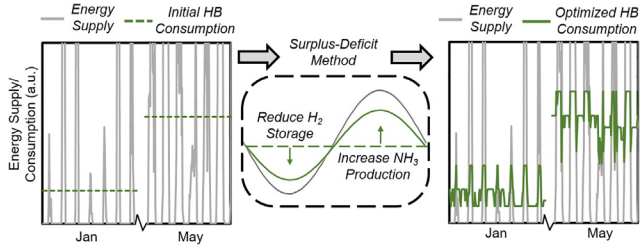


Figure 7. Schematic of surplus-deficit method for solving the variable Haber-Bosch process

The ammonia production level (i.e., energy consumption) is initialized (dashed green lines) with constant monthly ammonia production corresponding to 70%–80% of the average energy supply (solid gray lines) for a given month. Optimization occurs through increasing production at periods of surplus energy to increase overall ammonia production and decreasing production at periods of deficit to decrease the hydrogen storage requirement until optimal production scheduling is reached (solid green lines).

nitrogen) production is also adjusted up or down according to the corresponding buffer storage and final cost (detailed methodology in supplemental information). As a result, the method directly balances the two intuitive objectives for ammonia production with HB: to maximize overall production and minimize buffer storage to approach the minimum cost of ammonia as a function of ammonia and hydrogen production capacity and scheduling. By using this intuitive approach to solving the system, the limits of the HB ammonia production process under different renewable energy input conditions are clearly evidenced.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.joule.2023.12.002.

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AUTHOR CONTRIBUTIONS

Conceptualization, C.S. and L.T.-M.; methodology, C.S.; investigation, C.S.; writing – original draft, C.S.; writing – review and editing, C.S. and L.T.-M.; project administration, L.T.-M.

DECLARATION OF INTERESTS

The authors declare no competing interests.

INCLUSION AND DIVERSITY

We support inclusive, diverse, and equitable conduct of research.

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Supplemental information

The importance of dynamic operation and renewable energy source on the economic feasibility of green ammonia

Collin Smith and Laura Torrente-Murciano

Supplemental Figures

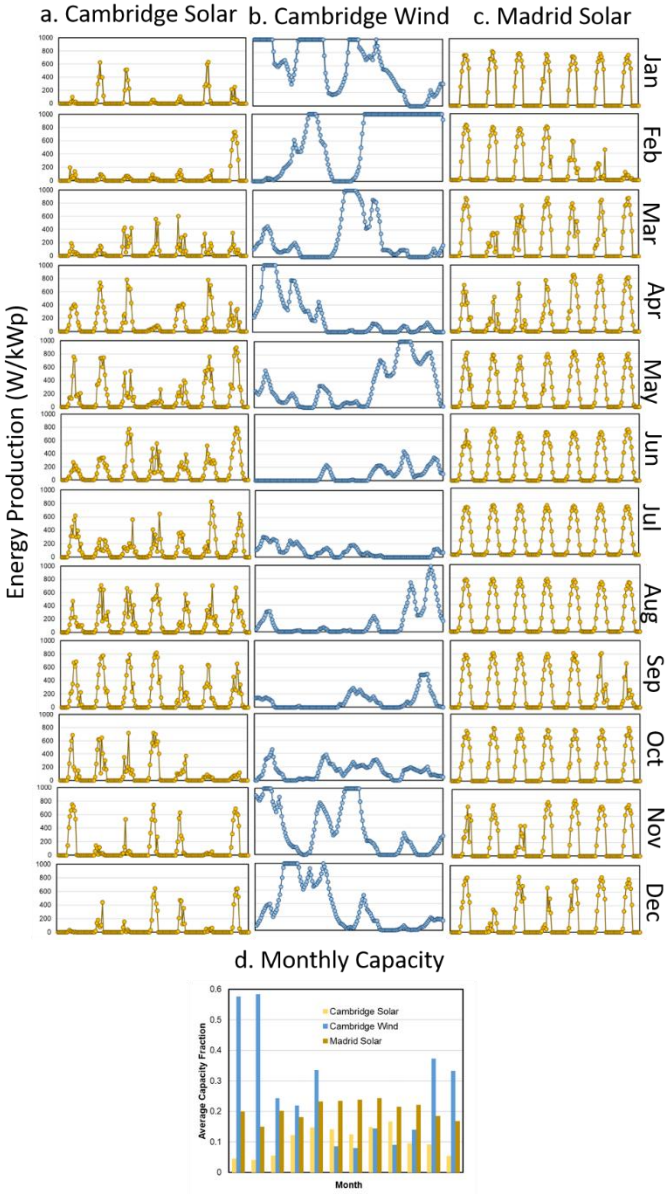


Figure S.1 | Representative weekly energy supply profiles per kWp of installed capacity for solar energy in Cambridge (a), wind energy in Cambridge (b), and solar energy in Madrid (c). d. Average monthly capacity.

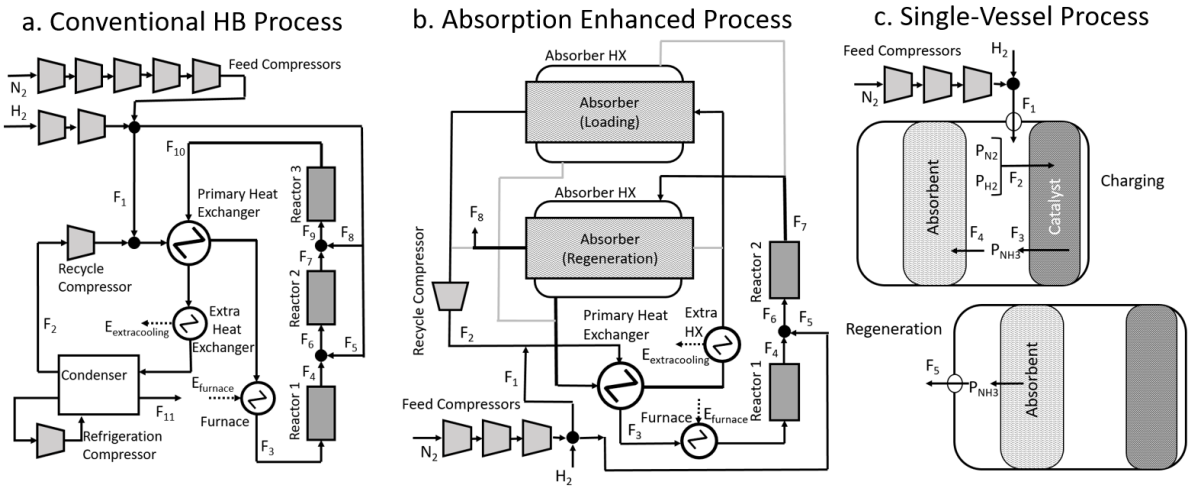


Figure S.2 | Detailed schematics of ammonia production processes. The conventional HB process requires a chain of feed compressors due to high pressures and separation is conducted with a condenser. The absorption enhanced process operates at 30 bar with absorption for ammonia separation and only two reactors lead to equilibrium. The single-vessel process also operates at 30 bar with the absorbent under the same conditions as the catalyst. The process flow labels are those used in the process simulations.

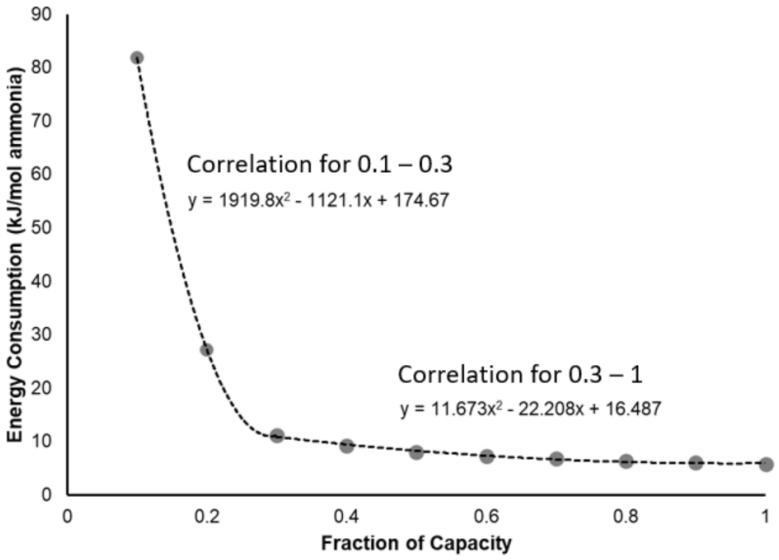


Figure S.3 | Energy consumption for variable production conventional HB process using N_2/H_2 ratio as the ramping method. The gradual increase as fraction of capacity decreases is a result of recycle and refrigeration not being as effective. The sharp increase in energy consumption at very low fraction of capacity is a result of heat required to keep the inlet of the reactor at 400°C. Two correlation curves were utilized to capture these two regimes for further analyses.

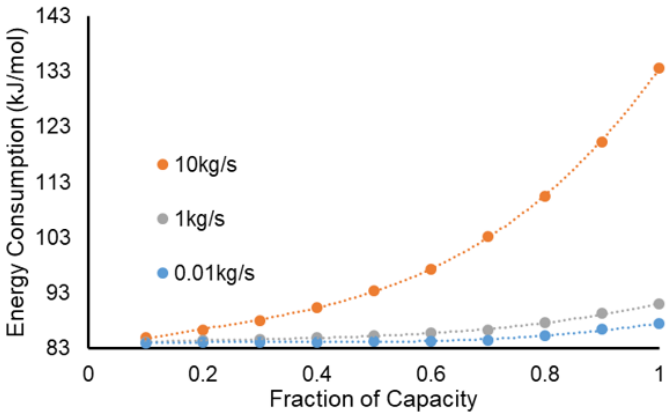


Figure S.4 | Energy consumption of absorption enhanced process as a function of capacity fraction and design capacity. At lower fraction capacity and design capacity, energy consumption decreases due to decreased pressure drop through the absorbers, until it reaches the limit of energy required for the regeneration of the absorber.

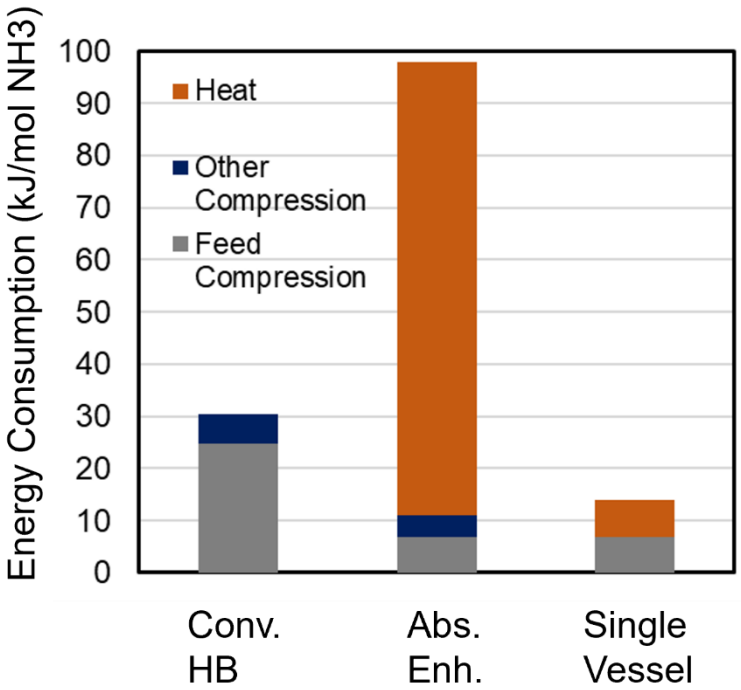


Figure S.5 | Comparison of energy consumption at full capacity 1 kg/s for each ammonia production process. The conventional HB process does not require heat input, but requires more feed compression due to high-pressures, in addition to recycle and refrigeration compression (i.e. other). The absorption enhanced and single-vessel processes require heat for regeneration of the absorbent, with the single-vessel process requiring significantly less heat due to the ease with which heat can be integrated in the simplified process without a recycle.

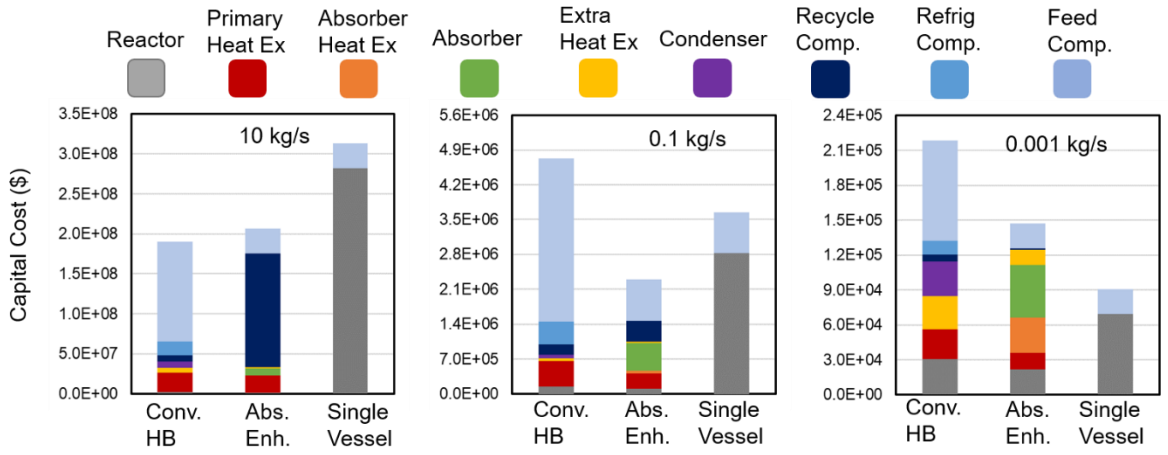


Figure S.6 | Comparison of capital costs for conventional HB, absorption enhanced, and single vessel production processes at three design capacities. Capital costs were calculated by using process simulations to size equipment and input size factors into capital estimates (see detailed methodology).

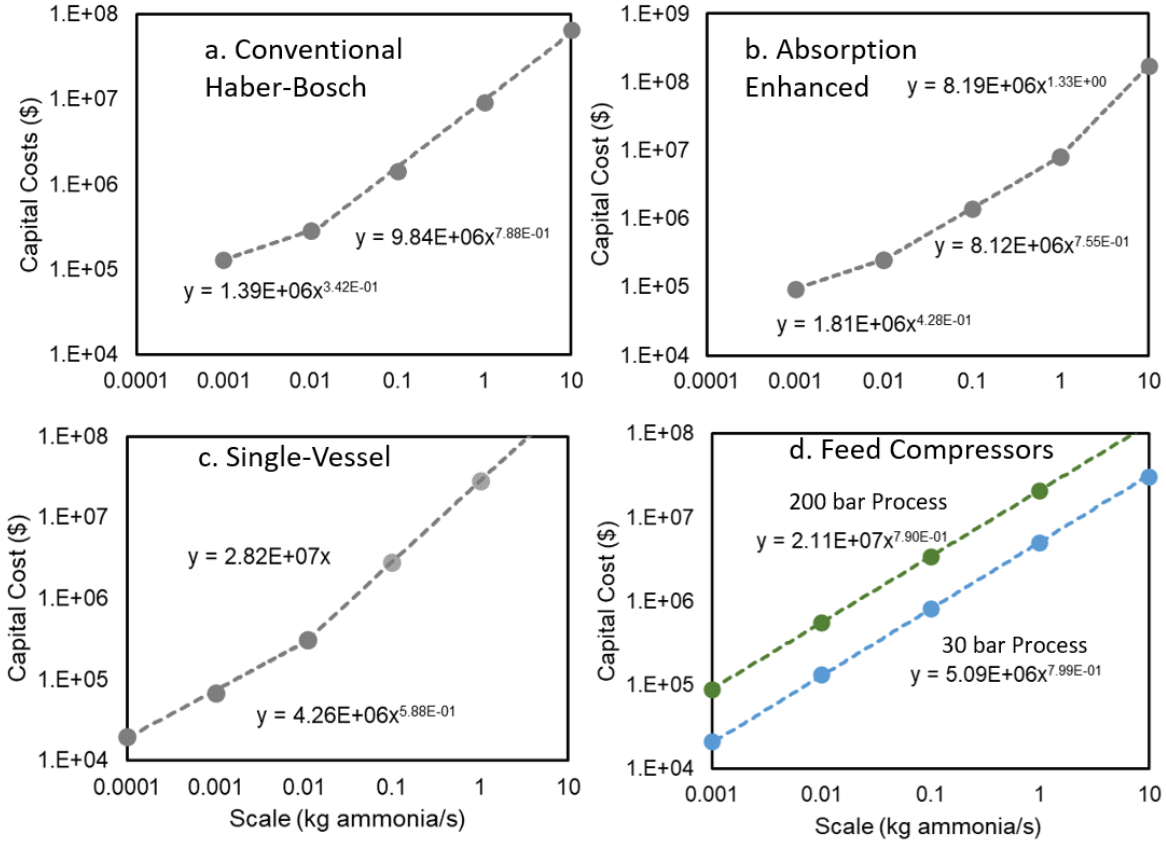


Figure S.7 | Capital cost correlations from compiled equipment costs for a. conventional HB, b. absorption enhanced, and c. single-vessel processes and d. feed compressors. The feed compressors where separated from the other costs so that they could be sized independently for buffer storage.

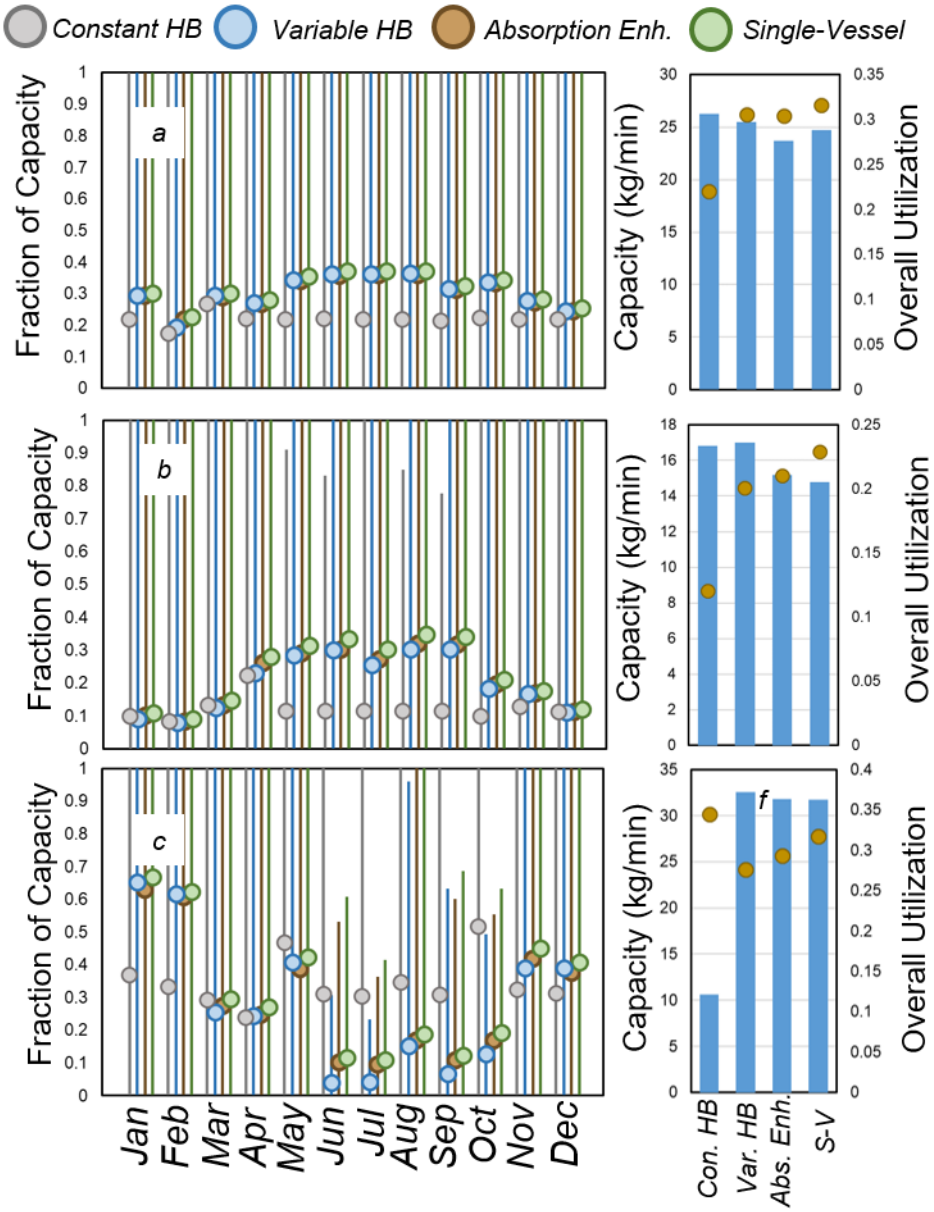


Figure S.8 | Dynamics of Hydrogen/Nitrogen production over the year constant HB, variable HB, absorption-enhanced, and single-vessel processes. a-c Fractional monthly capacity for solar energy in Madrid (a), solar energy in Cambridge (b), and wind energy in Cambridge (c). d-f Capacity of hydrogen production (blue bars) and overall yearly utilization (yellow circles)..

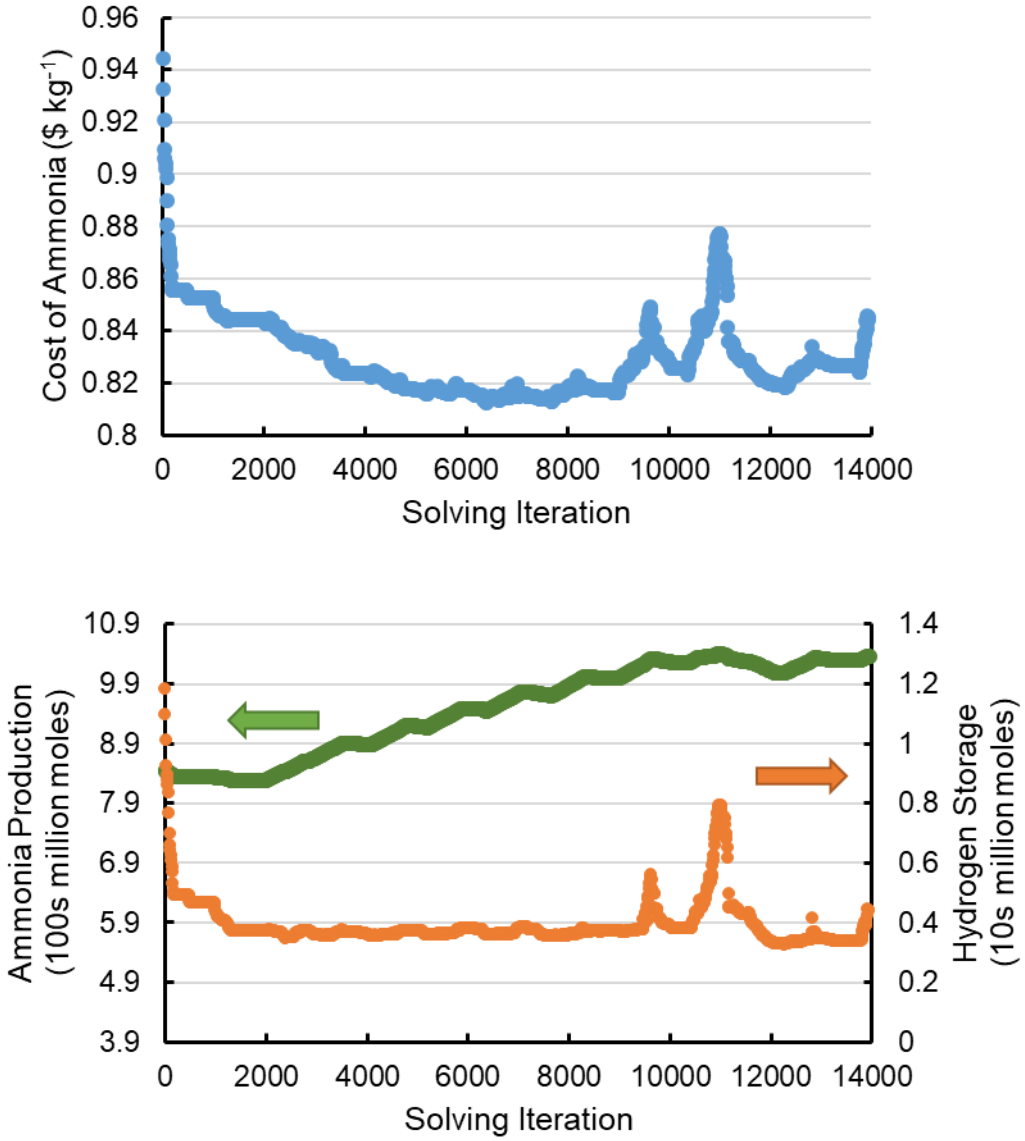


Figure S.9 | Output of the Surplus-Deficit solving method as a function of iteration in the algorithm. Displayed is the specific case of 100 MW of solar energy in Madrid. a. overall cost of ammonia. b. overall amount of ammonia production (green) and amount of H_2 buffer storage (orange).

Supplemental Tables

Table S.1| Number of Loop "Cycles" Between Production States

Time (min) [cycles]	**Final Production State (% capacity)**									
Initial State (%)	**100**	**90**	**80**	**70**	**60**	**50**	**40**	**30**	**20**	**10**
100	-	30 [2]	30 [2]	30 [2]	45 [3]	45 [3]	60 [4]	60 [4]	75 [5]	105 [7]
90	30 [2]	-	15 [1]	15 [1]	30 [2]	30 [2]	45 [3]	45 [3]	60 [4]	90 [6]
80	45 [3]	30 [2]	-	15 [1]	30 [2]	30 [2]	30 [2]	45 [3]	60 [4]	75 [5]
70	60 [4]	45 [3]	30 [2]	-	15 [1]	15 [1]	30 [2]	30 [2]	45 [3]	75 [5]
60	90 [6]	75 [5]	45 [3]	30 [2]	-	15 [1]	15 [1]	30 [2]	45 [3]	75 [5]
50	120 [8]	105 [7]	75 [5]	60 [4]	30 [2]	-	15 [1]	30 [2]	45 [3]	60 [4]
40	150 [10]	135 [9]	105 [7]	90 [6]	60 [4]	45 [3]	-	15 [1]	30 [2]	60 [4]
30	180 [12]	165 [11]	150 [10]	135 [9]	105 [7]	75 [5]	45 [3]	-	15 [1]	45 [3]
20	240 [16]	225 [15]	210 [14]	180 [12]	165 [11]	135 [9]	105 [7]	60 [4]	-	30 [2]
10	360 [24]	345 [23]	315 [21]	300 [20]	270 [18]	240 [16]	210 [14]	180 [12]	120 [8]	-

Table S.2 | Energy requirements for conventional HB at percent production capacity for a design capacity of 1 kg s^{-1}.

	100%	90%	80%	70%	60%	50%	40%	30%	20%	10%
Recycle (kW)	53	54	56	57	58	60	61	63	64	65
Furnace (kW)	-1064	-943	-815	-679	-529	-372	-208	-34	145	328
Extra heat (kW)	961	877	794	720	649	586	527	474	425	381
Condenser compressor (kW)	190	176	159	143	127	111	94	78	62	46
Feed Compression (kW)	1507	1363	1211	1062	908	757	607	455	302	151
Area HX (m^2)	898									
Area HX extra(m^2)	146									
Area Condenser(m^2)	191									

Table S.3 | Energy requirements for the absorption enhanced process at percent production capacity for a design capacity of 1 kg s^{-1}.

	100%	90%	80%	70%	60%	50%	40%	30%	20%	10%
Recycle (kW)	844	620	448	316	215	139	84	44	19	4
Absorber heat (kW)	2227	1964	1709	1471	1246	1038	834	626	417	209
Furnace (kW)	2892	2599	2305	2007	1724	1436	1146	859	572	286
Feed Compression (kW)	290	261	232	203	174	145	116	87	58	29
Area HX Absorber (m^2)	417									
Area HX (m^2)	1946									
Area HX extra (m^2)	131									

Table S.4 | Capital cost relationships for major ammonia production process equipment.

Equipment	Relationship	Source
Compressor	$C_{reciprocating} = F_M F_P F_{L\&M} C_{FOB}$ Where: $C_{FOB} = 180000 * \left(\frac{P[kW]}{100}\right)^{0.79}$ $F_m \sim 2.5\ for\ stainless\ steel$ $F_p = 1.3\ for\ high\ pressures$ $F_{L\&M} = ancillary\ labor\ and\ materials \sim 2.5$	[1]
Reactors (vertical vessel)	$C_{BM} = C_v * F_M * F_{BM}$ Where: $C_v[\$] = e^{5.6336+0.4599*\ln(W)+0.00582*\ln(W^2)}$ $F_M = material\ factor \sim 2\ for\ Stainless\ Steel$ $F_{BM} = bare\ module\ factor \sim 3$ $W[lbs] = \pi(D_i + t_p)(L + 0.8 * D_i)t_p \rho$ $t_p[in] = \frac{P_d D_i}{2 * SE - 1.2 * P_d}$ $\rho\left[\frac{lb}{in^3}\right] = 0.284$ $P_d[psig] = internal\ design\ pressure = 1.1 * operating\ pressure$ $D_i[in] = internal\ diameter$ $S[psi] = \max allowable\ stree = 13100psi$ $E = fraction\ weld\ efficiency = 1$	[2]
Absorbers (horizontal vessel)	$C_{BM} = C_v * F_M * F_{BM}$ Where: $C_v[\$] = e^{5.6336+0.4599*\ln(W)+0.00582*\ln(W^2)}$ [all else same as for reactors]	[2]
Heat Exchanger (shell & tube)	$C_{HXST} = F_{BM} F_P F_M F_L C_B$ Where: $C_B = exp^{11.551-0.9186*\ln(A[ft^2])+0.09005*\ln(A[ft^2]^2)}$ $F_M = material\ factor = 2.7 + \left(\frac{A[ft^2]}{100}\right)^{0.07}$ $F_L = length\ factor \sim 1.25$ $F_P = pressure\ factor = 0.9803 + 0.018\left(\frac{P[psi]}{100}\right) + 0.0017\left(\frac{P[psi]}{100}\right)^2$ $F_{BM} \sim 2$	[2]
Heat Exchanger (double pipe)	$C_{HXDP} = F_{BM} F_P F_M C_B$ Where: $C_B = exp^{7.2718-0.16*\ln(A[ft^2])}$ $F_M = material\ factor \sim 3\ for\ Stainless\ Steel$ $F_P = pressure\ factor = 0.851 + 0.1292\left(\frac{P[psi]}{600}\right) + 0.0198\left(\frac{P[psi]}{600}\right)^2$ $F_{BM} \sim 2$	[2]

Table S.5 | Capital, Energy, and Operating Costs for key pieces of equipment.

Parameter	Value	Source
Solar Panel Capital*	0.42 M€ MWp^{-1} at large scale (MWs) 0.8 M€ MWp^{-1} at small scale (kWs)	3
Solar Panel O&M Cost*	7,000 € MWp^{-1} y^{-1} at large scale (MWs) 10,400 € MWp^{-1} y^{-1} at small scale (kWs)	3
Wind Turbine Capital*	1.1 M€ MWp^{-1} at large scale (MWs) 3.8 M€ MWp^{-1} at small scale (kWs)	3
Wind Turbine O&M Cost*	14,000 € MWp^{-1} y^{-1} at large scale (MWs) 95,000 € MWp^{-1} y^{-1} at small scale (kWs)	3
Electrolyser Capital**	\$700 kW^{-1} at 100 MW \$1200 kW^{-1} at 100 kW	4
Electrolyser O&M Cost	2% of capital cost per year	5
N_2 PSA Cost	\$33,000(Capacity [kg hr^{-1}] /10)$^{0.61}$	6
Cost of gas storage	€800 kg_{H2}^{-1} (Hydrogen) €60 kg_{N2}^{-1} (Nitrogen)	7
Cost of battery storage	\$150 kWh^{-1}	7
Electrolyser Energy Consumption	379 kJ mol_{H2}^{-1} (75% HHV; 63% LHV)	5,8
N_2 PSA Energy Consumption	35 kJ mol_{N2}^{-1}	9,10

*Euros converted to dollars with a conversion factor of 1.18 \$ $€^{-1}$

**At intermediate scale, cost interpolated between large and small scales

Table S.6 | Algorithm steps for solving the “surplus-deficit” method

Step 1		When searching for the greatest deficit, restrict the search range to periods of the year with overall energy deficit, as determined through the H_2 balance over the year. In other words, examine before the minimum and after the maximum of the H_2 balance.
Step 2		Target the point of greatest deficit in the energy balance within the search range and decrease the ammonia production at that time by 10%.
Step 3		Check that the scheduling does not violate constraints of ramping, adjust neighboring time points as necessary. If production at the target time point is below the minimum production level, drop to zero production and extend the shutdown period to subsequent points for a period of time (typically 3 days).
Step 4		Solve for the new hydrogen and battery storage
	a	If the hydrogen production capacity does not allow the net hydrogen balance to be >0 for the year, increase the hydrogen capacity such that the balance is zero.
	b	Starting from the previous H_2 production capacity, decrease the capacity incrementally.
	c	Solve for the change in H_2 and battery inventory at every time over the year constrained by the maximum H_2 production capacity (remaining energy to batteries).
	d	Solve for the required H_2 and battery storage by adjusting the maximum inventory until the net balance for the year is zero (i.e. trim off peaks).
	e	If the adjustment in H_2 production capacity decreases cost, repeat steps 4b-4d.
Step 5		Keep the new ammonia production profile with down-shifted deficit point if either i) the cost has decreased or ii) the required H_2 storage has decreased AND the required energy for ammonia production does not drop below the energy supply.
Step 6*		Repeat steps 1-5 2000 times before moving to next step. After every iteration, the previous target point of deficit is ignored until all other points have been considered.
Step 7		Target the point of greatest surplus energy balance and increase ammonia production by 10%.
Step 8		Check that the scheduling does not violate constraints of ramping, adjust neighboring points as necessary. If the production capacity has increased, all time-points operating at minimum production level (typically 10%) are necessarily shifted up.
Step 9		Solve for the new hydrogen and battery storage
	a	If the hydrogen production capacity does not allow the net hydrogen balance to be >0 for the year, increase the hydrogen capacity such that the balance is zero.
	b	Starting from the previous H_2 production capacity, increase the capacity incrementally.
	c	Solve for the change in H_2 and battery inventory at every time over the year constrained by the maximum H_2 production capacity (remaining energy to batteries).
	d	Solve for the required H_2 and battery storage by adjusting the maximum inventory until the net balance for the year is zero (i.e. trim off peaks).
	e	If the adjustment in H_2 production capacity reduced the cost, repeat steps 9b -9d.
Step 10		Keep the new ammonia production profile with surplus point adjusted in all cases.
Step 11		Repeat steps 1 – 10 (skipping Step 6) a given number of times or until the cost exceeds the minimum by 15%.

*Time-points with deficit are specifically targeted first before increasing production at time-points with surplus energy because increasing production capacity would also increase minimum production capacity before downward adjustments of ammonia production have reached the original minimum.

Supplemental Methodology

To conduct an economic analysis of the ammonia production methods in the context of both the cost of ammonia from specific energy sources, the formulation was subdivided into i) process modeling, ii) capital cost estimation and iii) optimization with a renewable energy supply for ammonia production.

Process Modelling

i. Conventional Haber-Bosch Process

The conventional Haber-Bosch (HB) process was modelled according to the configuration shown in Figure S.2. While there are many potential reactor, feed, and separation configurations,[11] the set-up of three reactors with intercooling and feed after recompression was used as a model configuration [This is a simplification of the more complex configuration whereby recycle and feed provide intercooling]. The pressure of the process was approximated to be 200 bar, with a 10 bar pressure drop between the series of reactors and the condenser. Process flows and temperatures in the process were calculated through mass and energy balances iteratively solved in MATLAB for a given production rate until steady state is reached, as defined by <0.01% change in recycle flow between iterations of solving the loop.

The flows and conversion through each of the three reactors was solved by discretizing them into small segments using the kinetic equation derived by Sehested for a conventional iron catalyst.[12] The inlet temperature of the first reactor was held constant at 400°C, and the inlet to the second and third reactors was determined by the cold feed intercooling, with a minimum temperature of 400°C and a maximum fresh feed flow determined by the target production rate. Each of the reactors was sized such that the outlet of the reactor either reaches the maximum temperature of 500°C due to the exothermic reaction, or otherwise reaches 90% of equilibrium conversion.

In order to maximize the conversion, it was determined to use all the inlet gases for intercooling of the reactors (F5 and F8), negating F1. Given heat losses in a non-ideal reactor system, this configuration it not likely to be necessary in an actual process (i.e. some inlet flow will be F1), but this will not have a large effect on the present economic analysis.

The ramping of the conventional Haber-Bosch process was simulated through the adjustment of the N_2/H_2 ratio, which had previously been shown to be the most stable method for ramping production and has been patented in a similar scheme by Casale where inerts are utilized rather than an increased N_2 concentration.[13,14] Simply changing the flow rate in the process loop is not a viable option because it will cause a runaway reaction that exceeds the maximum temperature of the reactor. To simulate a change in N_2/H_2 ratio, the inlet flows were set to pure nitrogen for a given set of “cycles” through the system, followed by a feed enriched with nitrogen for a given number of “cycles” before returning to the necessary stoichiometric ratio. The nitrogen concentration and number of “cycles” were repeatedly adjusted to achieve the targeted reduction in production rate. This facilitated the calculation of flows and compositions at reduced production rates and the number of “cycles” through the system to achieve that production rate (Table S.1). Similarly, for upward ramping, the inlet flows were set to pure hydrogen.

As shown in Table S.1, the process loop is able to move rapidly between production states. However, at very low capacities, it becomes dramatically slower because fresh gas can only be replaced at the rate that it is removed through production, which is necessarily reduced. Similarly, it is approximated that 10% production capacity is the reasonable minimum

because lower production rates would be very difficult to ramp up from. The “cycles” were translated to time between production rates by approximately sizing the internal volume of the main pieces of equipment. It was approximated that the density of the catalyst is 2.5 g cm^{-3}, the void fraction is 0.4, and the tube piping for heat exchangers was 1.5 cm. With an oversizing factor of 10X to account for dispersion, thermal mass effects, and additional dead volume, the cycle time was approximated to be ~15 min. This is in agreement with previously reported approximations that the process was ramped at a rate of ~20% capacity per hour, although it can be much slower at low levels of production.[15]

To calculate the energy consumption of the conventional HB process as a function of production capacity, the process was divided into compression energy for the feed, recycle, and condensation compressors, and heat energy to keep the temperature of the reactor inlet at 400°C. To start, compression was assumed to be isentropic, where the ideal energy required is defined by Sandler,[16] and the compression and drive efficiency is 75% and 95%, respectively. For the feed, compression was approximated to happen in factors of 3 such that there are 6 steps for nitrogen (from 1 bar to 300 bar) and 3 steps for hydrogen (from 30 bar to 300 bar) assuming the electrolyzer has an output pressure of 30 bar and all compressed synthesis gas must reach the pressure of the buffer storage. For the recycle compressor, the energy is calculated using a pressure drop from 200 to 190 bar. For the condenser, the compression energy was calculated using ammonia as the working fluid and estimating the compression energy needed per unit of cooling power using the pressure enthalpy diagram. It was assumed refrigeration provides the cooling from 25°C to 10°C and the heat of vaporization.

The outlet temperatures of the heat exchanger were calculated from a simple energy balance, with the area required for heat exchange using the log mean temperature difference and a heat transfer coefficient of 200W m^{-2} K^{-1}.[17] The extra energy required for the furnace was estimated from an energy balance with the assumption that the inlet of the first reactor must be 400°C. For all except low production capacities, the furnace energy was negative, indicating the extra energy must be simply discarded to ambient air before entering the reactor (or otherwise the reactor effluent is cooler before entering the heat exchanger).

Due to greater thermal mass exiting the third reactor than returning in the recycle, F10 must be cooled further with water in an extra heat exchanger to reach ambient temperature. The water pumping energy was discarded because it was small relative to compression. The heat exchange and area of the extra water cooled heat exchanger and condenser were calculated just as for the primary gas heat exchanger.

For production capacities less than 100%, the energy requirements, as shown in Table S.2, were calculated using the heat exchanger areas sized for full capacity, and assuming the recycle, feed, and condenser compressors have full flexibility. For the sake of simplicity during simulations, the energy consumption per ammonia produced for the process loop (excluding feed) was fit to a calibration curve (Figure S.3). The feed compression energy was excluded because it correlates directly to the production rate. Heat losses during start–up and shut-down were neglected from the assessment.

ii. Absorption Enhanced Process

The schematic of the process used to simulate an absorption enhanced process is shown in Figure S.2. Principally, the condenser is replaced with an absorber that can remove ammonia to a lower pressure, facilitating the usage of a lower pressure process loop. In particular, a process pressure of 30 bar was chosen because it is the readily achievable

pressure for electrolysis, and it was approximated that the absorbent can remove ammonia completely. For the purposes of calculating flows, the loop was approximated to have no pressure drop, which would not be the case with the packed absorber bed (energy calculations below). But, since it is approximated that the absorber removes all the ammonia, the pressure would not have an effect.

The reactor system with intercooling was sized according the same procedure as the conventional HB process. However, because of the greatly reduced pressure, the absorption enhanced process reactors reach >90% equilibrium before the maximum temperature, and thereby only require one stage of intercooling to marginally increase the conversion after the first reactor. As such, the final reactor effluent is F7. As in the conventional HB process, it was optimal to have all inlet flow used for intercooling.

The recycle rate was chosen as the method for throttling the production rate. As opposed to the conventional reactor at 200 bar, the reactor at 30 bar reaches equilibrium before the maximum temperature, and therefore will not induce a runaway reaction. In practice, this method would be implemented by adjusting the recycle compressor and carefully controlling the inlet flow to keep pressure in the system constant. As a result, even though this method can rapidly change the production rate, and was approximated to be instantaneous (on an hourly time scale), it will require precise control. Further, although completely stopping production will cause problems with heat integration, it is possible by stopping recycle.

Similar to the conventional HB process, the energy consumption for the absorption enhanced process was calculated by dividing it into compression and heat energy. Feed compression was divided into three stages for nitrogen, while hydrogen required no compression because the system is at 30 bar. Recycle compression required sizing the absorbers.

To size the absorbers, the amount of absorbent required (in this case $CaCl_2$) was calculated by approximating a 30 min cycle time for the absorbers and the best case capacity of 1 mole ammonia per mole calcium chloride. The volume and length of the absorber was then calculated with an L/D of 6 and assuming the absorbent is supported on silica, as is common practice.[18] With the viscosity, density, and velocity of gas in the absorber (using inlet composition for viscosity and density), the pressure drop in the packed bed was calculated from the Ergun equation [16] by approximating a void fraction of 0.35, a particle size of 0.5 mm, and a constant velocity. From this, the compression energy was calculated as before.

The amount of heat energy required for the regeneration of the absorber from an absorption temperature of 50°C to a regeneration temperature of 300°C was calculated using the heat capacity of the absorber and combining with the enthalpy of desorption. The energy exchanged to the absorber by the effluent of the second reaction (F7) was calculated by assuming the final temperature of F7 as 10°C higher than the desorption temperature. The area of the absorber heat exchanger was then calculated by approximating the absorber temperature as constant at the desorption temperature (i.e. most heat is used for desorption), and the extra heat needed to input into the absorber simply calculated as the difference between that needed and that supplied by the reactor effluent. Since the heat exchanged with the absorber only partially cools the reactor effluent, a primary gas heat exchange with the recycle gases is still required just as with the conventional HB process, whereby the final temperatures are solved such that the temperature differential is always >10°C. Due to the lower starting temperature (310°C) for the inlet of the high temperature stream, the recycled gases must be heated in a furnace to reach 400°C for the reactor inlet. Conversely, the outlet of the high temperature stream must undergo additional water cooling before the absorber,

which was calculated as for the conventional HB, and was similarly a minimal contribution to energy and capital costs.

Just as for the conventional HB process, the energy requirements for the absorption enhanced process at production rates below the design capacity were calculated using the areas for heat exchange sized at full capacity and approximating the compressors to achieve perfect ramping. As shown in Table S.3, the energy required for absorber regeneration and reheating gases dominates the energy requirements. However, because the recycle compression is strongly dependent on the absolute size of the absorber, the energy requirement was calculated as a function of design production capacity and used to construct a series of correlation curves, as shown in Figure S.4. As expected, the energy consumption per ammonia produced decreases both at lower fraction capacities and also at lower design capacities due to reduced flow rate through the absorber. However, the decrease in energy consumption reaches a limit due the heat for regeneration.

iii. Single-Vessel Process

The single-vessel process was sized according to the basic design criteria that the catalytic rate must match the absorption rate, as governed by kinetic equations developed previously for a 5%Ru/10%Cs/CeO_2 catalyst and 50%$MnCl_2$/SiO_2 absorbent with an operating pressure of 30 bar and temperature of ~330°C.[19] The mass ratio of absorbent to catalyst was designed as 3:1 because this resulted in a cycle time of ~30 min. The desorption half of the semi-batch cycle was assumed to occur in the same amount of time as the absorption half of the semi-batch cycle, resulting in an overall reaction rate of 0.2 mmol g_{cat}^{-1} min^{-1} after the temperature, pressure, and N_2:H_2 ratio has been optimized. The heat for desorption was approximated to be supplied through integration of heat generated in the first half of the semi-batch cycle. The process was designed in reactor modules with a maximum capacity of 0.01 kg s^{-1} because this resulted in a reaction vessel 0.9 m in diameter and 4.4 m in length. At capacities higher than this, the modules were repeated, while at lower capacities the modules were down-sized. For cost estimations, the price of ruthenium in the catalyst was included at \$9 g^{-1}. For the design of the process, it was assumed the vessel is well-mixed and no ammonia decomposition occurs over the catalyst during desorption.

Capital Cost Estimation

The capital costs for each of the ammonia production processes were estimated by considering reactors, absorbers, compressors, and heat exchangers as the major pieces of equipment. The estimates for the reactor, absorber, and heat exchanger capital costs were derived from [2]. For the reactors and absorbers, the volume was derived from the catalyst/absorbent mass and density. For the heat exchangers, the capital costs were divided between shell and tube heat exchangers, which were employed for areas >50 m^2, and double pipe heat exchangers, which were employed for areas <50 m^2. The compressors for all three processes were assumed to be reciprocating because they are superior at variable flow rates, as opposed to centrifugal compressors common in steady state processes. The capital cost estimates in this case were derived from [1] due to a wider range of input power for small scales. A summary of the capital cost relationships is given in Table S.4, including material, pressure, and bare module factors, etc. In order to adjust the capital relationships to the monetary value of the dollar in 2020 (CEPCI ~600), the compressor cost was multiplied by a factor of 1.15 (to adjust from a CEPCI of 522 in 2013) and the reactor, absorber, and heat exchanger cost was multiplied by 1.06 (to adjust from CEPCI of 567 in 2009). The monetary value of 2020 was

selected as the basis for analysis to avoid interferences related to the high inflation rates experienced in 2022-23.

For the sake of simplicity when conducting economic simulations with renewable energy profiles, the capital costs for each process were fit to calibrations curves rather than sizing the system for each simulation production rate. As shown in Figure S.7, for each of the processes, there are different calibration curves for regions of scale due to shifts in the limiting costs. Further, due to the necessity of sizing the feed compressors for the conventional HB processes at the same scale as the electrolysers and PSA for buffer gas storage, correlation curves were constructed for feed compression alone and all other parts of the processes.

The solar and wind energy supplies were derived directly from data provided by [20] for a fixed position solar panel with optimized orientation in the case of solar energy, and from the wind speed at 10 m extrapolated to 80 m and applied to a typical energy production curve for a wind turbine.[21] The cost and energy estimates for solar panels, wind turbines and gas generation and storage are shown in Table S.5. Due to the data these cost estimates were published, they are approximated to be at parity with the basis of 2020 dollars.

The capital cost for major pieces of equipment was annualized using a discount rate of 7% and a lifetime of 30 years. Operating and maintenance costs were included for solar panels, wind turbines, and electrolysers (as shown in Table S.5) because they constitute the largest costs in the process and have well-defined operating and maintenance costs. To calculate the cost of ammonia, the sum of annualized capital and O&M was divided by the total yearly ammonia production. For specifically calculating the cost of renewable electricity from solar and wind in each of the case studies, the sum of annualized energy capital and O&M was divided by the total yearly energy generation.

Optimization with renewable energy supply

The difficulty of simulating the implementation of ammonia production to a variable renewable energy supply stems principally from matching the transience constraints in ammonia production to the energy supply profile in an optimal manner. As a result, ammonia production at every time, the amount of buffer storage, and the capacity of hydrogen and nitrogen production are independent variables when simulating the conventional HB process. A method termed “surplus-deficit” was implemented in this analysis for solving the capacity and scheduling of the HB process. In short, this method starts with an initial constant ammonia production rate, with points of “surplus” energy where the ammonia production rate should be increased and points of “deficit” energy where the ammonia production rate should be decreased. Incremental movement of ammonia production up and down at individual points in search of minimum cost will eventually generate an approximate optimization. Within this method, the renewable energy supply for every hour over a year was simplified by using a representative week repeated either four or five times to simulate a month (shown in Figure S.1).

A steps of the program executed mathematically in MATLAB are shown in Table S.6. To start, with a given renewable energy supply, the ammonia production profile is initialized to be constant in each month and equal to some fraction of the total energy supplied in that month. Typically, this fraction was 80% for solar energy but 70% for wind energy due to a greater fluctuation in supply between seasons, however, the initial value did not have a large effect on the final costs (<5%). If this initial constant ammonia production was less than the lower limit of production capacity (typically 10%, but also varied), the constant ammonia production was reassigned to the production capacity minimum. With this initial profile for ammonia

production, the program then enters the surplus deficit method which is repeated a given number of times or until the cost has increased 15% from the minimum. An example of the iterative outputs for this method are shown in Figure S.9. The cost of ammonia forms a “U” shape with a minimum as a function of iteration as the algorithm simultaneously decreases hydrogen storage and increases total ammonia production until after many iterations the design capacity for ammonia production is excessively high and is constraining the downward ramp of the process.

In the case of constant ammonia production using the conventional HB process (no ramping over the year), the cost was calculated by simply increasing the ammonia production rate incrementally, solving for hydrogen and battery storage through the hydrogen production capacity as in steps (8a)-(8e), and calculating the total cost. In every simulation, it was optimal to oversize the hydrogen production capacity such that it would be possible to make more hydrogen than necessary over the year as rapid replenishment of the hydrogen buffer decreases the necessary buffer size.

Similarly, in the case of perfectly flexible (time scale of an hour) ammonia production with either the absorption enhanced or single-vessel processes, the yearly ammonia production profile was determined by incrementally increasing the production capacity until reaching the minimum cost. At every time point, ammonia production would follow energy supply up to the maximum production. Hydrogen and battery buffers were not required, and brief calculations indicated that distributing ammonia production over period within energy would be detrimental due to the cost of compressors and storage even while the capital costs of the ammonia production decrease.

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